# project1-Copy2

March 17, 2025

## 1 Section 1: Import Libraries and Setup

[1]:
```
import os
import cv2 as cv2

import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import tensorflow as tf


from tensorflow.keras import layers, models
from sklearn.model_selection import train_test_split
from tensorflow.keras.preprocessing import image_dataset_from_directory
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras import layers, regularizers
from tensorflow.keras.optimizers.schedules import ExponentialDecay
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping
from sklearn.utils.class_weight import compute_class_weight
```

[8]:
```
if tf.config.list_physical_devices('GPU'):
    print("GPU is available.")
else:
    print("GPU is NOT available.")
```

```
GPU is available.
```

[2]:
```
train_datagen = ImageDataGenerator(
    rotation_range=40,
    width_shift_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
)
```

[3]:
```
# Define the dataset path on Kaggle
path = 'D:/files/LUND/DeepLearning/project/pokemon-dataset-1000/'
dataset_path = path + 'dataset'
```

```
train_path = path + 'train'
test_path = path + 'test'
val_path = path + 'val'
```

[6]:
```
def generate_train_labels(Train_path, save_path="train_labels.npy"):
    #
    class_names = sorted([name for name in os.listdir(Train_path) if os.path.
 ↪isdir(os.path.join(Train_path, name))])
    class_to_idx = {name: idx for idx, name in enumerate(class_names)}

    #
    train_labels = []
    for class_name in class_names:
        class_dir = os.path.join(Train_path, class_name)
        for img_file in os.listdir(class_dir):
            img_path = os.path.join(class_dir, img_file)
            if os.path.isfile(img_path):
                train_labels.append(class_to_idx[class_name])

    #   .npy
    train_labels = np.array(train_labels, dtype=np.int32)
    np.save(save_path, train_labels)
    print(f"    {save_path}  {len(train_labels)}   ")

#
generate_train_labels(Train_path=train_path,save_path=path+'train_labels.npy')
```

```
    D:/files/LUND/DeepLearning/project/pokemon-
dataset-1000/train_labels.npy  19434
```

[7]:
```
train_dataset = train_datagen.flow_from_directory(
    train_path,
    target_size=(224, 224),  #
    batch_size=128,
    class_mode='categorical',  #
    seed=42
)
```

```
Found 19434 images belonging to 1000 classes.
```

[8]:
```
val_dataset = image_dataset_from_directory(
    val_path,
    label_mode="categorical",
    image_size=(224, 224),
    seed=42
)
```

```
Found 2815 files belonging to 1000 classes.
```

[9]:
```
test_dataset = image_dataset_from_directory(
    test_path,
    label_mode="categorical",
    image_size=(224, 224),
    seed=42
)
```

```
Found 4290 files belonging to 1000 classes.
```

[10]:
```
train_labels = np.load("train_labels.npy")
class_weights = compute_class_weight('balanced', classes=np.
 ↪unique(train_labels), y=train_labels)
class_weight_dict = dict(enumerate(class_weights))
```

## 2 ResNet1 Learning Rate 0.001 Adam

[12]:
```
#  ResNet
base_model = tf.keras.applications.ResNet50(
    weights='imagenet',
    include_top=False,
    input_shape=(224, 224, 3)
)
base_model.trainable = False  #

model_resnet = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(1024, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(1000, activation='softmax')
])
lr_schedule = ExponentialDecay(initial_learning_rate=0.001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = Adam(learning_rate=lr_schedule)
```

[13]:
```
model_resnet.compile(
    optimizer=optimizer,
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

#   epochs=20
```

[14]:
```
history_resnet = model_resnet.fit(
```

```
    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict
)
```

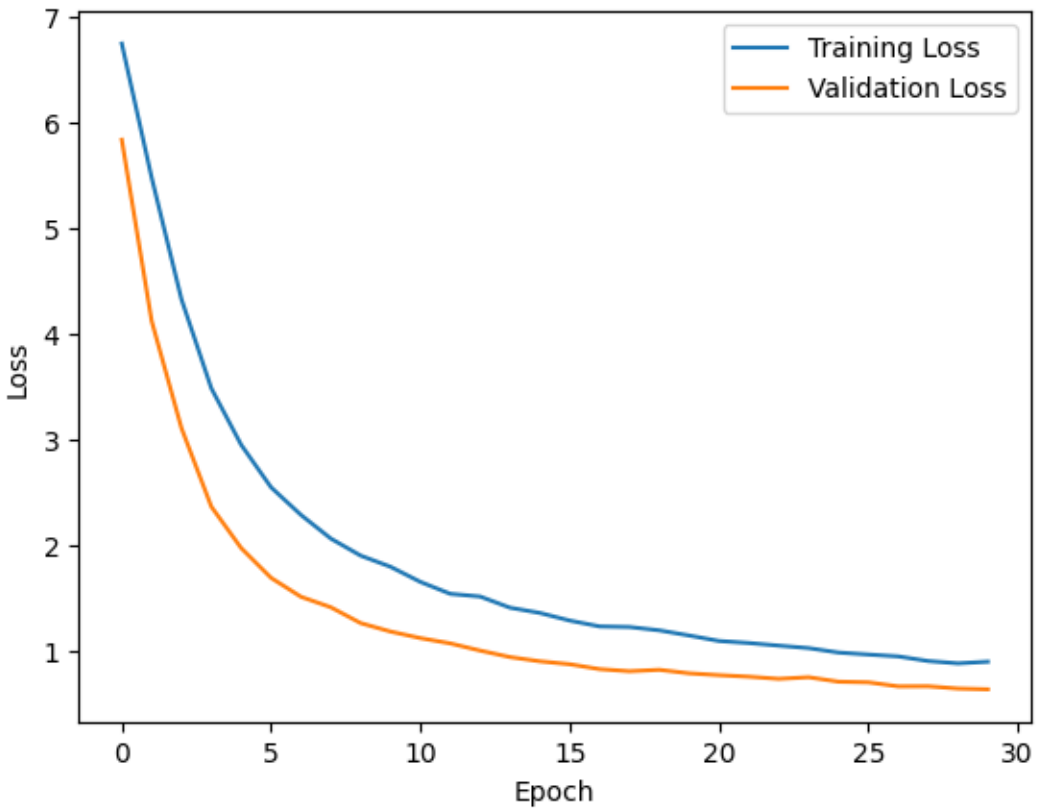
```
Epoch 1/30
152/152 [==============================] - 236s 2s/step - loss: 7.1231 -
accuracy: 0.0053 - val_loss: 5.8307 - val_accuracy: 0.0721
Epoch 2/30
152/152 [==============================] - 123s 805ms/step - loss: 5.7587 -
accuracy: 0.0635 - val_loss: 4.1152 - val_accuracy: 0.3012
Epoch 3/30
152/152 [==============================] - 123s 810ms/step - loss: 4.4174 -
accuracy: 0.1633 - val_loss: 3.1000 - val_accuracy: 0.4174
Epoch 4/30
152/152 [==============================] - 123s 808ms/step - loss: 3.5511 -
accuracy: 0.2548 - val_loss: 2.3569 - val_accuracy: 0.5361
Epoch 5/30
152/152 [==============================] - 123s 809ms/step - loss: 2.9905 -
accuracy: 0.3183 - val_loss: 1.9651 - val_accuracy: 0.5933
Epoch 6/30
152/152 [==============================] - 123s 806ms/step - loss: 2.5880 -
accuracy: 0.3857 - val_loss: 1.6835 - val_accuracy: 0.6409
Epoch 7/30
152/152 [==============================] - 122s 803ms/step - loss: 2.2806 -
accuracy: 0.4264 - val_loss: 1.5046 - val_accuracy: 0.6718
Epoch 8/30
152/152 [==============================] - 123s 806ms/step - loss: 2.0315 -
accuracy: 0.4682 - val_loss: 1.4053 - val_accuracy: 0.6849
Epoch 9/30
152/152 [==============================] - 122s 803ms/step - loss: 1.8883 -
accuracy: 0.4853 - val_loss: 1.2553 - val_accuracy: 0.7076
Epoch 10/30
152/152 [==============================] - 123s 805ms/step - loss: 1.7726 -
accuracy: 0.5118 - val_loss: 1.1755 - val_accuracy: 0.7321
Epoch 11/30
152/152 [==============================] - 123s 806ms/step - loss: 1.6306 -
accuracy: 0.5449 - val_loss: 1.1134 - val_accuracy: 0.7460
Epoch 12/30
152/152 [==============================] - 123s 805ms/step - loss: 1.5398 -
accuracy: 0.5640 - val_loss: 1.0635 - val_accuracy: 0.7545
Epoch 13/30
152/152 [==============================] - 123s 810ms/step - loss: 1.4975 -
accuracy: 0.5713 - val_loss: 0.9953 - val_accuracy: 0.7655
Epoch 14/30
152/152 [==============================] - 123s 805ms/step - loss: 1.3999 -
accuracy: 0.5929 - val_loss: 0.9339 - val_accuracy: 0.7812
```

```
Epoch 15/30
152/152 [==============================] - 122s 804ms/step - loss: 1.3447 -
accuracy: 0.6091 - val_loss: 0.8939 - val_accuracy: 0.7872
Epoch 16/30
152/152 [==============================] - 123s 807ms/step - loss: 1.2376 -
accuracy: 0.6306 - val_loss: 0.8662 - val_accuracy: 0.7943
Epoch 17/30
152/152 [==============================] - 123s 807ms/step - loss: 1.2172 -
accuracy: 0.6336 - val_loss: 0.8198 - val_accuracy: 0.8082
Epoch 18/30
152/152 [==============================] - 123s 805ms/step - loss: 1.1913 -
accuracy: 0.6411 - val_loss: 0.8002 - val_accuracy: 0.8075
Epoch 19/30
152/152 [==============================] - 123s 805ms/step - loss: 1.1601 -
accuracy: 0.6447 - val_loss: 0.8134 - val_accuracy: 0.8028
Epoch 20/30
152/152 [==============================] - 123s 805ms/step - loss: 1.1253 -
accuracy: 0.6598 - val_loss: 0.7801 - val_accuracy: 0.8075
Epoch 21/30
152/152 [==============================] - 123s 806ms/step - loss: 1.0817 -
accuracy: 0.6570 - val_loss: 0.7628 - val_accuracy: 0.8135
Epoch 22/30
152/152 [==============================] - 123s 807ms/step - loss: 1.0408 -
accuracy: 0.6728 - val_loss: 0.7483 - val_accuracy: 0.8249
Epoch 23/30
152/152 [==============================] - 123s 808ms/step - loss: 1.0395 -
accuracy: 0.6759 - val_loss: 0.7279 - val_accuracy: 0.8238
Epoch 24/30
152/152 [==============================] - 123s 807ms/step - loss: 0.9846 -
accuracy: 0.6873 - val_loss: 0.7432 - val_accuracy: 0.8224
Epoch 25/30
152/152 [==============================] - 123s 807ms/step - loss: 0.9730 -
accuracy: 0.6953 - val_loss: 0.7008 - val_accuracy: 0.8298
Epoch 26/30
152/152 [==============================] - 123s 808ms/step - loss: 0.9455 -
accuracy: 0.6937 - val_loss: 0.6949 - val_accuracy: 0.8306
Epoch 27/30
152/152 [==============================] - 123s 808ms/step - loss: 0.9175 -
accuracy: 0.6964 - val_loss: 0.6566 - val_accuracy: 0.8341
Epoch 28/30
152/152 [==============================] - 123s 810ms/step - loss: 0.8840 -
accuracy: 0.7114 - val_loss: 0.6575 - val_accuracy: 0.8348
Epoch 29/30
152/152 [==============================] - 123s 808ms/step - loss: 0.8626 -
accuracy: 0.7188 - val_loss: 0.6359 - val_accuracy: 0.8416
Epoch 30/30
152/152 [==============================] - 123s 810ms/step - loss: 0.9028 -
accuracy: 0.7108 - val_loss: 0.6288 - val_accuracy: 0.8469
```

[15]:
```
test_loss, test_acc = model_resnet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 30s 211ms/step - loss: 0.6937 -
accuracy: 0.8373
Test Accuracy: 0.8373
```

[16]:
```
plt.plot(history_resnet.history['loss'], label='Training Loss')
plt.plot(history_resnet.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




# ResNet2 Learning Rate 0.0001 Adam

[18]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.0001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = Adam(learning_rate=lr_schedule)
model_resnet.compile(
```

```
    optimizer=optimizer,
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

#   epochs=20
```

[19]:
```
history_resnet2 = model_resnet.fit(

    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict
)
```

```
Epoch 1/30
152/152 [==============================] - 125s 811ms/step - loss: 0.7351 -
accuracy: 0.7548 - val_loss: 0.5854 - val_accuracy: 0.8572
Epoch 2/30
152/152 [==============================] - 123s 806ms/step - loss: 0.7187 -
accuracy: 0.7612 - val_loss: 0.5800 - val_accuracy: 0.8554
Epoch 3/30
152/152 [==============================] - 123s 806ms/step - loss: 0.6915 -
accuracy: 0.7667 - val_loss: 0.5810 - val_accuracy: 0.8579
Epoch 4/30
152/152 [==============================] - 123s 808ms/step - loss: 0.6774 -
accuracy: 0.7743 - val_loss: 0.5795 - val_accuracy: 0.8561
Epoch 5/30
152/152 [==============================] - 123s 809ms/step - loss: 0.6899 -
accuracy: 0.7743 - val_loss: 0.5704 - val_accuracy: 0.8572
Epoch 6/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6642 -
accuracy: 0.7740 - val_loss: 0.5674 - val_accuracy: 0.8572
Epoch 7/30
152/152 [==============================] - 123s 809ms/step - loss: 0.6484 -
accuracy: 0.7740 - val_loss: 0.5688 - val_accuracy: 0.8604
Epoch 8/30
152/152 [==============================] - 123s 809ms/step - loss: 0.6631 -
accuracy: 0.7757 - val_loss: 0.5736 - val_accuracy: 0.8586
Epoch 9/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6511 -
accuracy: 0.7787 - val_loss: 0.5669 - val_accuracy: 0.8597
Epoch 10/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6441 -
accuracy: 0.7861 - val_loss: 0.5661 - val_accuracy: 0.8600
Epoch 11/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6362 -
accuracy: 0.7885 - val_loss: 0.5623 - val_accuracy: 0.8604
```

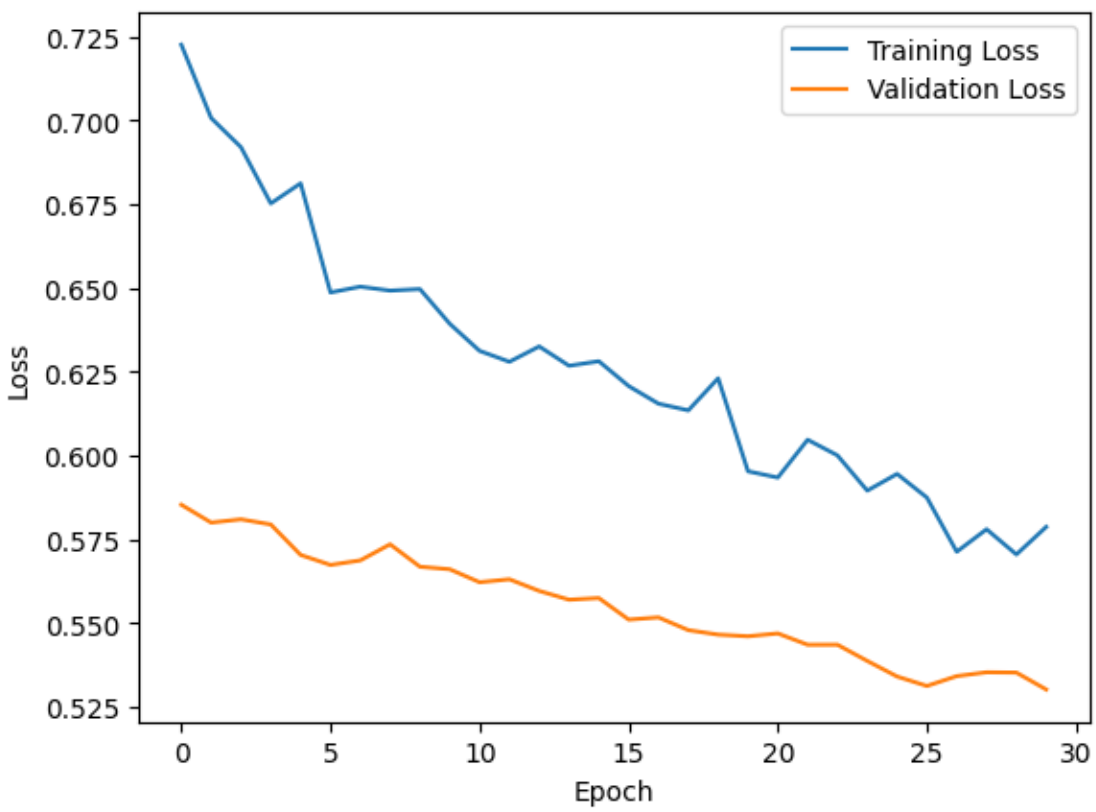
```
Epoch 12/30
152/152 [==============================] - 123s 808ms/step - loss: 0.6292 -
accuracy: 0.7860 - val_loss: 0.5631 - val_accuracy: 0.8611
Epoch 13/30
152/152 [==============================] - 123s 806ms/step - loss: 0.6248 -
accuracy: 0.7835 - val_loss: 0.5597 - val_accuracy: 0.8611
Epoch 14/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6226 -
accuracy: 0.7833 - val_loss: 0.5570 - val_accuracy: 0.8604
Epoch 15/30
152/152 [==============================] - 123s 806ms/step - loss: 0.6297 -
accuracy: 0.7832 - val_loss: 0.5575 - val_accuracy: 0.8607
Epoch 16/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6223 -
accuracy: 0.7900 - val_loss: 0.5511 - val_accuracy: 0.8650
Epoch 17/30
152/152 [==============================] - 123s 806ms/step - loss: 0.6224 -
accuracy: 0.7911 - val_loss: 0.5518 - val_accuracy: 0.8622
Epoch 18/30
152/152 [==============================] - 123s 806ms/step - loss: 0.6174 -
accuracy: 0.7910 - val_loss: 0.5480 - val_accuracy: 0.8632
Epoch 19/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6120 -
accuracy: 0.7971 - val_loss: 0.5466 - val_accuracy: 0.8639
Epoch 20/30
152/152 [==============================] - 124s 811ms/step - loss: 0.6020 -
accuracy: 0.7936 - val_loss: 0.5461 - val_accuracy: 0.8661
Epoch 21/30
152/152 [==============================] - 124s 813ms/step - loss: 0.5941 -
accuracy: 0.7983 - val_loss: 0.5470 - val_accuracy: 0.8657
Epoch 22/30
152/152 [==============================] - 123s 810ms/step - loss: 0.6052 -
accuracy: 0.7944 - val_loss: 0.5436 - val_accuracy: 0.8636
Epoch 23/30
152/152 [==============================] - 124s 812ms/step - loss: 0.6137 -
accuracy: 0.7902 - val_loss: 0.5436 - val_accuracy: 0.8668
Epoch 24/30
152/152 [==============================] - 124s 814ms/step - loss: 0.6037 -
accuracy: 0.7951 - val_loss: 0.5388 - val_accuracy: 0.8679
Epoch 25/30
152/152 [==============================] - 123s 810ms/step - loss: 0.5769 -
accuracy: 0.7989 - val_loss: 0.5341 - val_accuracy: 0.8664
Epoch 26/30
152/152 [==============================] - 126s 826ms/step - loss: 0.5864 -
accuracy: 0.7975 - val_loss: 0.5313 - val_accuracy: 0.8689
Epoch 27/30
152/152 [==============================] - 126s 830ms/step - loss: 0.5788 -
accuracy: 0.8012 - val_loss: 0.5343 - val_accuracy: 0.8682
```

```
Epoch 28/30
152/152 [==============================] - 126s 826ms/step - loss: 0.5618 -
accuracy: 0.8059 - val_loss: 0.5354 - val_accuracy: 0.8675
Epoch 29/30
152/152 [==============================] - 126s 830ms/step - loss: 0.5662 -
accuracy: 0.8047 - val_loss: 0.5353 - val_accuracy: 0.8664
Epoch 30/30
152/152 [==============================] - 126s 825ms/step - loss: 0.5870 -
accuracy: 0.8007 - val_loss: 0.5303 - val_accuracy: 0.8700
```

[20]:
```
test_loss, test_acc = model_resnet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 12s 87ms/step - loss: 0.6092 -
accuracy: 0.8636
Test Accuracy: 0.8636
```

[21]:
```
plt.plot(history_resnet2.history['loss'], label='Training Loss')
plt.plot(history_resnet2.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```



# 3 ResNet3 Learning Rate 0.0001 Nadam

[23]:
```
optimizer = tf.keras.optimizers.Nadam(
    learning_rate=0.0001,
    beta_1=0.9,  #
    beta_2=0.999  #
)
model_resnet.compile(
    optimizer=optimizer,
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

#   epochs=20
```

[24]:
```
history_resnet3 = model_resnet.fit(

    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict
)
```

```
Epoch 1/30
152/152 [==============================] - 128s 828ms/step - loss: 0.7102 -
accuracy: 0.7641 - val_loss: 0.6063 - val_accuracy: 0.8512
Epoch 2/30
152/152 [==============================] - 126s 827ms/step - loss: 0.7855 -
accuracy: 0.7507 - val_loss: 0.6130 - val_accuracy: 0.8512
Epoch 3/30
152/152 [==============================] - 126s 828ms/step - loss: 0.7709 -
accuracy: 0.7413 - val_loss: 0.5914 - val_accuracy: 0.8551
Epoch 4/30
152/152 [==============================] - 126s 826ms/step - loss: 0.7670 -
accuracy: 0.7463 - val_loss: 0.5948 - val_accuracy: 0.8579
Epoch 5/30
152/152 [==============================] - 126s 826ms/step - loss: 0.7555 -
accuracy: 0.7526 - val_loss: 0.5999 - val_accuracy: 0.8529
Epoch 6/30
152/152 [==============================] - 126s 825ms/step - loss: 0.7491 -
accuracy: 0.7534 - val_loss: 0.5947 - val_accuracy: 0.8519
Epoch 7/30
152/152 [==============================] - 126s 827ms/step - loss: 0.7104 -
accuracy: 0.7592 - val_loss: 0.5684 - val_accuracy: 0.8625
Epoch 8/30
```

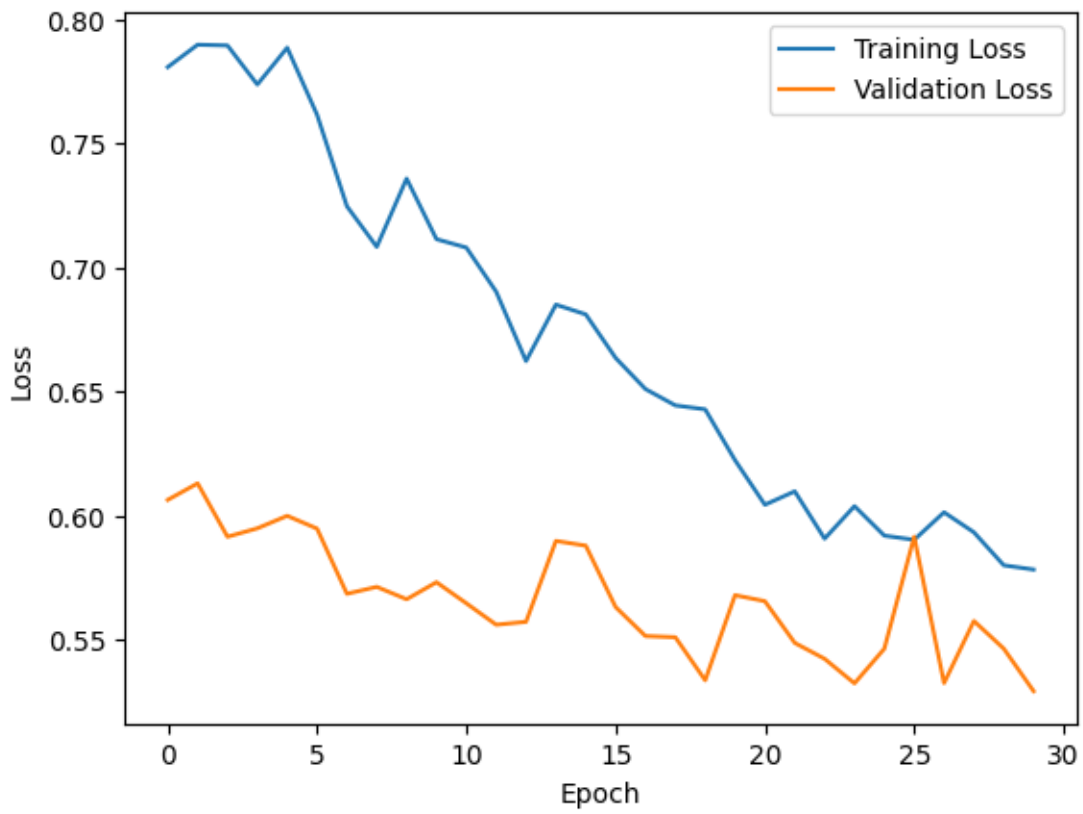
```
152/152 [==============================] - 125s 820ms/step - loss: 0.6872 -
accuracy: 0.7695 - val_loss: 0.5712 - val_accuracy: 0.8622
Epoch 9/30
152/152 [==============================] - 123s 809ms/step - loss: 0.7411 -
accuracy: 0.7579 - val_loss: 0.5662 - val_accuracy: 0.8622
Epoch 10/30
152/152 [==============================] - 124s 811ms/step - loss: 0.7022 -
accuracy: 0.7600 - val_loss: 0.5731 - val_accuracy: 0.8618
Epoch 11/30
152/152 [==============================] - 123s 811ms/step - loss: 0.7071 -
accuracy: 0.7602 - val_loss: 0.5646 - val_accuracy: 0.8625
Epoch 12/30
152/152 [==============================] - 123s 810ms/step - loss: 0.6669 -
accuracy: 0.7730 - val_loss: 0.5559 - val_accuracy: 0.8668
Epoch 13/30
152/152 [==============================] - 123s 810ms/step - loss: 0.6431 -
accuracy: 0.7757 - val_loss: 0.5571 - val_accuracy: 0.8636
Epoch 14/30
152/152 [==============================] - 123s 810ms/step - loss: 0.6618 -
accuracy: 0.7737 - val_loss: 0.5897 - val_accuracy: 0.8615
Epoch 15/30
152/152 [==============================] - 123s 809ms/step - loss: 0.6644 -
accuracy: 0.7739 - val_loss: 0.5878 - val_accuracy: 0.8629
Epoch 16/30
152/152 [==============================] - 123s 809ms/step - loss: 0.6556 -
accuracy: 0.7751 - val_loss: 0.5631 - val_accuracy: 0.8661
Epoch 17/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6441 -
accuracy: 0.7758 - val_loss: 0.5514 - val_accuracy: 0.8675
Epoch 18/30
152/152 [==============================] - 123s 806ms/step - loss: 0.6375 -
accuracy: 0.7788 - val_loss: 0.5509 - val_accuracy: 0.8675
Epoch 19/30
152/152 [==============================] - 123s 808ms/step - loss: 0.6234 -
accuracy: 0.7850 - val_loss: 0.5335 - val_accuracy: 0.8657
Epoch 20/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5895 -
accuracy: 0.7900 - val_loss: 0.5679 - val_accuracy: 0.8689
Epoch 21/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6113 -
accuracy: 0.7902 - val_loss: 0.5654 - val_accuracy: 0.8661
Epoch 22/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6011 -
accuracy: 0.7888 - val_loss: 0.5486 - val_accuracy: 0.8710
Epoch 23/30
152/152 [==============================] - 123s 805ms/step - loss: 0.5988 -
accuracy: 0.7917 - val_loss: 0.5422 - val_accuracy: 0.8696
Epoch 24/30
```

```
152/152 [==============================] - 122s 804ms/step - loss: 0.6044 -
accuracy: 0.7931 - val_loss: 0.5323 - val_accuracy: 0.8718
Epoch 25/30
152/152 [==============================] - 123s 805ms/step - loss: 0.5763 -
accuracy: 0.8015 - val_loss: 0.5463 - val_accuracy: 0.8710
Epoch 26/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5668 -
accuracy: 0.8009 - val_loss: 0.5915 - val_accuracy: 0.8639
Epoch 27/30
152/152 [==============================] - 123s 807ms/step - loss: 0.6060 -
accuracy: 0.7913 - val_loss: 0.5324 - val_accuracy: 0.8707
Epoch 28/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5655 -
accuracy: 0.8020 - val_loss: 0.5575 - val_accuracy: 0.8675
Epoch 29/30
152/152 [==============================] - 123s 811ms/step - loss: 0.5659 -
accuracy: 0.8012 - val_loss: 0.5463 - val_accuracy: 0.8750
Epoch 30/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5589 -
accuracy: 0.8042 - val_loss: 0.5291 - val_accuracy: 0.8764
```

[25]:
```
test_loss, test_acc = model_resnet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 12s 86ms/step - loss: 0.6164 -
accuracy: 0.8737
Test Accuracy: 0.8737
```

[26]:
```
plt.plot(history_resnet3.history['loss'], label='Training Loss')
plt.plot(history_resnet3.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




## 4 ResNet4 Learning Rate 0.0001 SGD

[28]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.0001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = tf.keras.optimizers.SGD(learning_rate=lr_schedule, momentum=0.9)
```

[29]:
```
history_resnet4 = model_resnet.fit(

    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict
)
```

```
Epoch 1/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5874 -
accuracy: 0.7989 - val_loss: 0.5460 - val_accuracy: 0.8725
Epoch 2/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5898 -
```

```
accuracy: 0.7998 - val_loss: 0.5427 - val_accuracy: 0.8689
Epoch 3/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5752 -
accuracy: 0.7997 - val_loss: 0.5336 - val_accuracy: 0.8703
Epoch 4/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5573 -
accuracy: 0.8048 - val_loss: 0.5363 - val_accuracy: 0.8806
Epoch 5/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5374 -
accuracy: 0.8099 - val_loss: 0.5447 - val_accuracy: 0.8742
Epoch 6/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5379 -
accuracy: 0.8096 - val_loss: 0.5388 - val_accuracy: 0.8753
Epoch 7/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5732 -
accuracy: 0.8063 - val_loss: 0.5426 - val_accuracy: 0.8774
Epoch 8/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5454 -
accuracy: 0.8111 - val_loss: 0.5334 - val_accuracy: 0.8817
Epoch 9/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5365 -
accuracy: 0.8134 - val_loss: 0.5307 - val_accuracy: 0.8817
Epoch 10/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5585 -
accuracy: 0.8112 - val_loss: 0.5131 - val_accuracy: 0.8753
Epoch 11/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5481 -
accuracy: 0.8111 - val_loss: 0.5436 - val_accuracy: 0.8817
Epoch 12/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5515 -
accuracy: 0.8103 - val_loss: 0.5551 - val_accuracy: 0.8732
Epoch 13/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5406 -
accuracy: 0.8132 - val_loss: 0.5429 - val_accuracy: 0.8767
Epoch 14/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5595 -
accuracy: 0.8092 - val_loss: 0.5453 - val_accuracy: 0.8750
Epoch 15/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5309 -
accuracy: 0.8125 - val_loss: 0.5400 - val_accuracy: 0.8785
Epoch 16/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5252 -
accuracy: 0.8181 - val_loss: 0.5251 - val_accuracy: 0.8813
Epoch 17/30
152/152 [==============================] - 123s 808ms/step - loss: 0.5027 -
accuracy: 0.8213 - val_loss: 0.5418 - val_accuracy: 0.8813
Epoch 18/30
152/152 [==============================] - 123s 806ms/step - loss: 0.4873 -
```

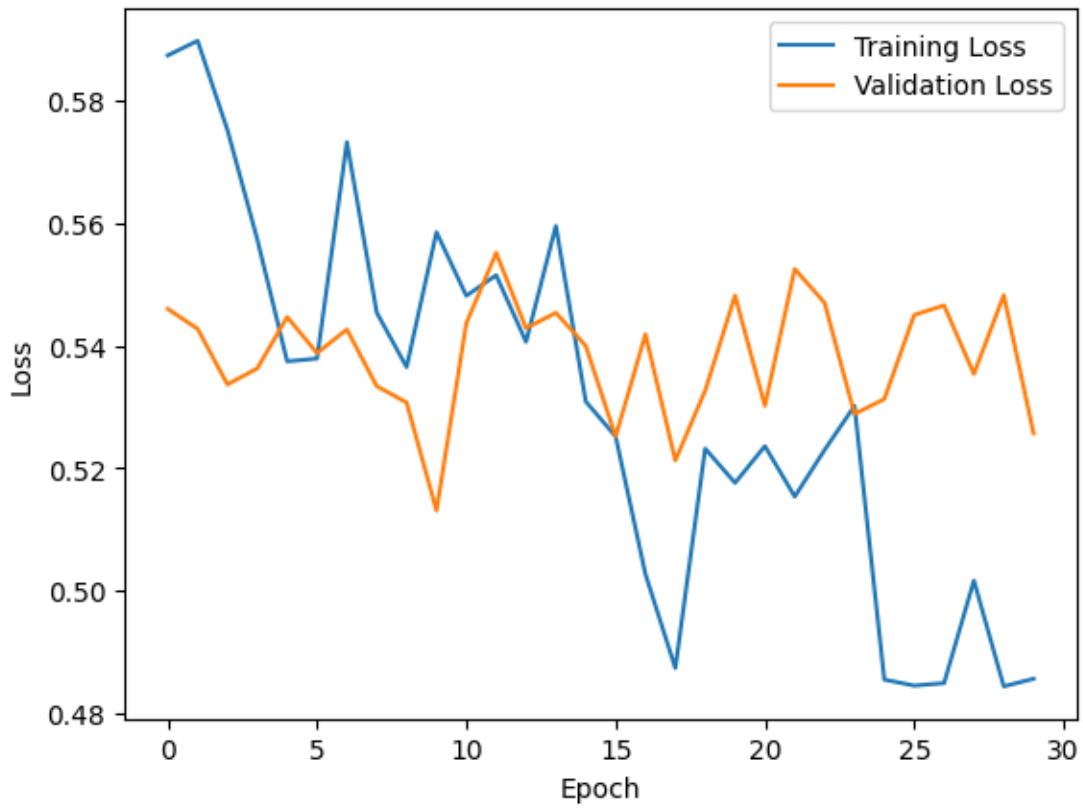
```
accuracy: 0.8280 - val_loss: 0.5212 - val_accuracy: 0.8821
Epoch 19/30
152/152 [==============================] - 123s 805ms/step - loss: 0.5232 -
accuracy: 0.8185 - val_loss: 0.5326 - val_accuracy: 0.8817
Epoch 20/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5176 -
accuracy: 0.8228 - val_loss: 0.5482 - val_accuracy: 0.8838
Epoch 21/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5236 -
accuracy: 0.8204 - val_loss: 0.5302 - val_accuracy: 0.8813
Epoch 22/30
152/152 [==============================] - 123s 806ms/step - loss: 0.5153 -
accuracy: 0.8207 - val_loss: 0.5525 - val_accuracy: 0.8785
Epoch 23/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5230 -
accuracy: 0.8184 - val_loss: 0.5470 - val_accuracy: 0.8817
Epoch 24/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5302 -
accuracy: 0.8177 - val_loss: 0.5288 - val_accuracy: 0.8845
Epoch 25/30
152/152 [==============================] - 123s 808ms/step - loss: 0.4855 -
accuracy: 0.8292 - val_loss: 0.5313 - val_accuracy: 0.8892
Epoch 26/30
152/152 [==============================] - 123s 808ms/step - loss: 0.4845 -
accuracy: 0.8317 - val_loss: 0.5450 - val_accuracy: 0.8828
Epoch 27/30
152/152 [==============================] - 123s 810ms/step - loss: 0.4849 -
accuracy: 0.8292 - val_loss: 0.5466 - val_accuracy: 0.8913
Epoch 28/30
152/152 [==============================] - 123s 807ms/step - loss: 0.5016 -
accuracy: 0.8259 - val_loss: 0.5354 - val_accuracy: 0.8824
Epoch 29/30
152/152 [==============================] - 123s 809ms/step - loss: 0.4844 -
accuracy: 0.8319 - val_loss: 0.5482 - val_accuracy: 0.8835
Epoch 30/30
152/152 [==============================] - 123s 808ms/step - loss: 0.4856 -
accuracy: 0.8313 - val_loss: 0.5257 - val_accuracy: 0.8920
```

[30]:
```
test_loss, test_acc = model_resnet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 12s 86ms/step - loss: 0.6724 -
accuracy: 0.8718
Test Accuracy: 0.8718
```

[31]:
```
plt.plot(history_resnet4.history['loss'], label='Training Loss')
plt.plot(history_resnet4.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
```

```
plt.ylabel('Loss')
plt.legend()
plt.show()
```




## 5 MoblieNet1 learning rate=0.001 Adam

[38]:
```
base_model = tf.keras.applications.MobileNetV2(input_shape=(224, 224, 3),
                          include_top=False,
                          weights='imagenet')
base_model.trainable = False

model_mobilenet = models.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(512, activation='relu', kernel_regularizer=regularizers.l2(0.
 ↪001)),
    layers.BatchNormalization(),
    layers.Dropout(0.5),
    layers.Dense(256, activation='relu', kernel_regularizer=regularizers.l2(0.
 ↪001)),
```

```
    layers.Dropout(0.3),
    layers.Dense(1000, activation='softmax')
])

lr_schedule = ExponentialDecay(initial_learning_rate=0.001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = Adam(learning_rate=lr_schedule)
model_mobilenet.compile(optimizer=optimizer, loss='categorical_crossentropy',␣
 ↪metrics=['accuracy'])

early_stop = EarlyStopping(monitor='val_accuracy', patience=5,␣
 ↪restore_best_weights=True)

#   epochs=15
```

[40]:
```
history_mobilenet1 = model_mobilenet.fit(
    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict,
    callbacks=[early_stop]
)
```

```
Epoch 1/30
152/152 [==============================] - 124s 804ms/step - loss: 7.9917 -
accuracy: 0.0041 - val_loss: 6.7633 - val_accuracy: 0.0394
Epoch 2/30
152/152 [==============================] - 121s 793ms/step - loss: 6.7572 -
accuracy: 0.0325 - val_loss: 5.4009 - val_accuracy: 0.1346
Epoch 3/30
152/152 [==============================] - 121s 796ms/step - loss: 5.6221 -
accuracy: 0.0811 - val_loss: 4.6947 - val_accuracy: 0.2004
Epoch 4/30
152/152 [==============================] - 120s 789ms/step - loss: 4.9127 -
accuracy: 0.1330 - val_loss: 4.1702 - val_accuracy: 0.2657
Epoch 5/30
152/152 [==============================] - 120s 790ms/step - loss: 4.4831 -
accuracy: 0.1721 - val_loss: 3.7687 - val_accuracy: 0.3162
Epoch 6/30
152/152 [==============================] - 120s 788ms/step - loss: 4.1959 -
accuracy: 0.2027 - val_loss: 3.5926 - val_accuracy: 0.3496
Epoch 7/30
152/152 [==============================] - 120s 790ms/step - loss: 3.9855 -
accuracy: 0.2322 - val_loss: 3.4160 - val_accuracy: 0.3922
Epoch 8/30
152/152 [==============================] - 120s 790ms/step - loss: 3.8633 -
```

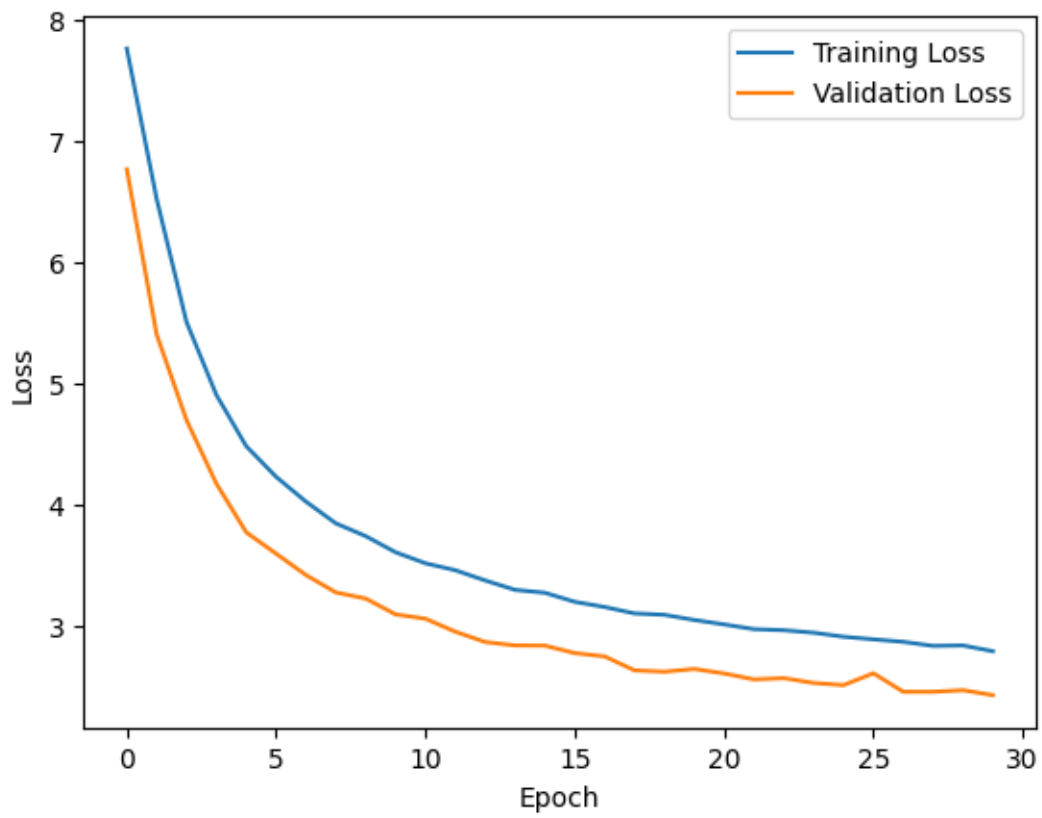
```
accuracy: 0.2482 - val_loss: 3.2728 - val_accuracy: 0.4089
Epoch 9/30
152/152 [==============================] - 120s 790ms/step - loss: 3.6901 -
accuracy: 0.2704 - val_loss: 3.2223 - val_accuracy: 0.4316
Epoch 10/30
152/152 [==============================] - 120s 790ms/step - loss: 3.5776 -
accuracy: 0.2809 - val_loss: 3.0915 - val_accuracy: 0.4554
Epoch 11/30
152/152 [==============================] - 120s 789ms/step - loss: 3.4908 -
accuracy: 0.2994 - val_loss: 3.0561 - val_accuracy: 0.4671
Epoch 12/30
152/152 [==============================] - 120s 787ms/step - loss: 3.4122 -
accuracy: 0.3172 - val_loss: 2.9487 - val_accuracy: 0.4721
Epoch 13/30
152/152 [==============================] - 120s 789ms/step - loss: 3.3585 -
accuracy: 0.3192 - val_loss: 2.8637 - val_accuracy: 0.4977
Epoch 14/30
152/152 [==============================] - 120s 789ms/step - loss: 3.2140 -
accuracy: 0.3308 - val_loss: 2.8355 - val_accuracy: 0.5012
Epoch 15/30
152/152 [==============================] - 120s 791ms/step - loss: 3.1983 -
accuracy: 0.3464 - val_loss: 2.8340 - val_accuracy: 0.4995
Epoch 16/30
152/152 [==============================] - 120s 790ms/step - loss: 3.1566 -
accuracy: 0.3526 - val_loss: 2.7721 - val_accuracy: 0.5041
Epoch 17/30
152/152 [==============================] - 120s 787ms/step - loss: 3.1187 -
accuracy: 0.3519 - val_loss: 2.7450 - val_accuracy: 0.5211
Epoch 18/30
152/152 [==============================] - 120s 789ms/step - loss: 3.0943 -
accuracy: 0.3591 - val_loss: 2.6299 - val_accuracy: 0.5382
Epoch 19/30
152/152 [==============================] - 120s 788ms/step - loss: 3.0482 -
accuracy: 0.3651 - val_loss: 2.6186 - val_accuracy: 0.5353
Epoch 20/30
152/152 [==============================] - 120s 788ms/step - loss: 2.9908 -
accuracy: 0.3763 - val_loss: 2.6414 - val_accuracy: 0.5449
Epoch 21/30
152/152 [==============================] - 120s 788ms/step - loss: 2.9745 -
accuracy: 0.3798 - val_loss: 2.6041 - val_accuracy: 0.5439
Epoch 22/30
152/152 [==============================] - 120s 788ms/step - loss: 2.9902 -
accuracy: 0.3736 - val_loss: 2.5552 - val_accuracy: 0.5481
Epoch 23/30
152/152 [==============================] - 120s 788ms/step - loss: 2.8832 -
accuracy: 0.3830 - val_loss: 2.5656 - val_accuracy: 0.5577
Epoch 24/30
152/152 [==============================] - 120s 791ms/step - loss: 2.9400 -
```

```
accuracy: 0.3821 - val_loss: 2.5257 - val_accuracy: 0.5549
Epoch 25/30
152/152 [==============================] - 120s 788ms/step - loss: 2.8847 -
accuracy: 0.3836 - val_loss: 2.5079 - val_accuracy: 0.5556
Epoch 26/30
152/152 [==============================] - 120s 792ms/step - loss: 2.8277 -
accuracy: 0.4001 - val_loss: 2.6060 - val_accuracy: 0.5460
Epoch 27/30
152/152 [==============================] - 120s 791ms/step - loss: 2.8520 -
accuracy: 0.3929 - val_loss: 2.4530 - val_accuracy: 0.5659
Epoch 28/30
152/152 [==============================] - 120s 791ms/step - loss: 2.8029 -
accuracy: 0.4093 - val_loss: 2.4528 - val_accuracy: 0.5773
Epoch 29/30
152/152 [==============================] - 120s 791ms/step - loss: 2.8088 -
accuracy: 0.4028 - val_loss: 2.4662 - val_accuracy: 0.5655
Epoch 30/30
152/152 [==============================] - 120s 788ms/step - loss: 2.7534 -
accuracy: 0.4106 - val_loss: 2.4254 - val_accuracy: 0.5776
```

[41]:
```
test_loss, test_acc = model_mobilenet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 8s 58ms/step - loss: 2.4420 -
accuracy: 0.5739
Test Accuracy: 0.5739
```

[42]:
```
plt.plot(history_mobilenet1.history['loss'], label='Training Loss')
plt.plot(history_mobilenet1.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




## 6 MobileNet2 learning rate=0.0001 Adam

[44]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.0001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = Adam(learning_rate=lr_schedule)
model_mobilenet.compile(optimizer=optimizer, loss='categorical_crossentropy',␣
 ↪metrics=['accuracy'])

early_stop = EarlyStopping(monitor='val_accuracy', patience=5,␣
 ↪restore_best_weights=True)

#   epochs=15
```

[45]:
```
history_mobilenet2 = model_mobilenet.fit(
    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict,
```

```
    callbacks=[early_stop]
)
```

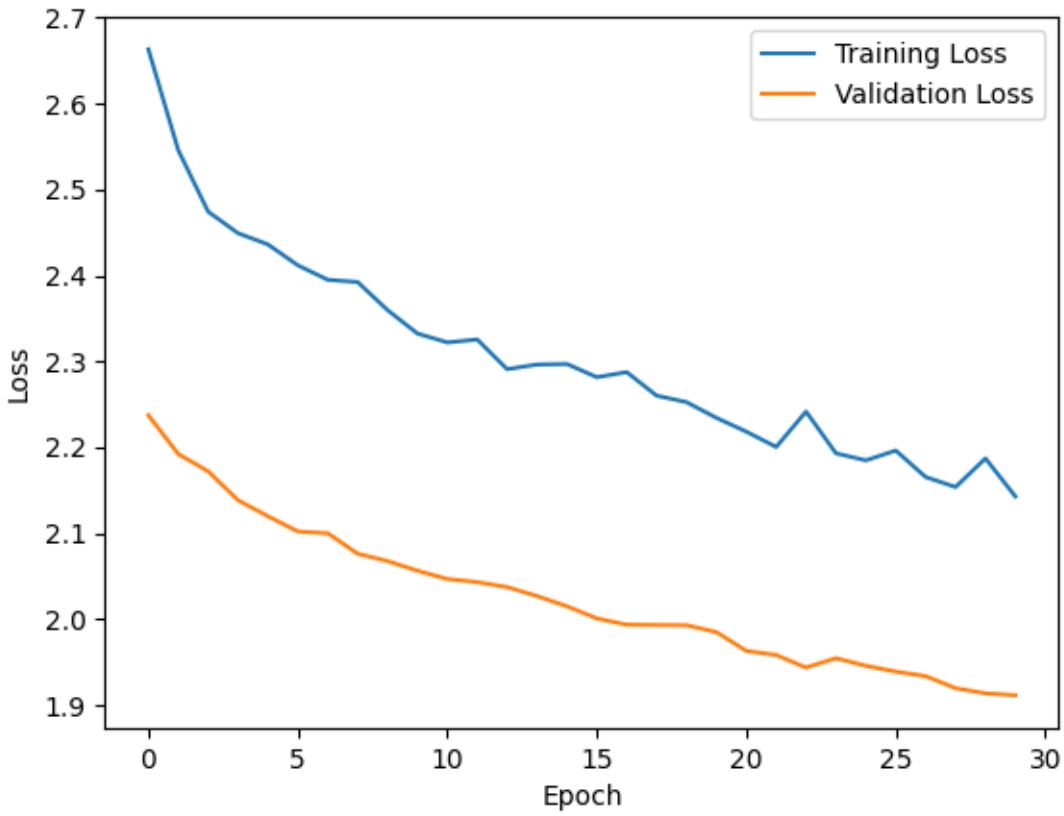
```
Epoch 1/30
152/152 [==============================] - 122s 796ms/step - loss: 2.6814 -
accuracy: 0.4342 - val_loss: 2.2373 - val_accuracy: 0.6188
Epoch 2/30
152/152 [==============================] - 121s 794ms/step - loss: 2.5433 -
accuracy: 0.4479 - val_loss: 2.1921 - val_accuracy: 0.6256
Epoch 3/30
152/152 [==============================] - 120s 792ms/step - loss: 2.4712 -
accuracy: 0.4619 - val_loss: 2.1720 - val_accuracy: 0.6309
Epoch 4/30
152/152 [==============================] - 120s 792ms/step - loss: 2.4336 -
accuracy: 0.4639 - val_loss: 2.1384 - val_accuracy: 0.6359
Epoch 5/30
152/152 [==============================] - 121s 794ms/step - loss: 2.4392 -
accuracy: 0.4686 - val_loss: 2.1196 - val_accuracy: 0.6373
Epoch 6/30
152/152 [==============================] - 120s 792ms/step - loss: 2.3994 -
accuracy: 0.4743 - val_loss: 2.1020 - val_accuracy: 0.6384
Epoch 7/30
152/152 [==============================] - 121s 792ms/step - loss: 2.3899 -
accuracy: 0.4792 - val_loss: 2.0999 - val_accuracy: 0.6387
Epoch 8/30
152/152 [==============================] - 120s 792ms/step - loss: 2.4031 -
accuracy: 0.4758 - val_loss: 2.0761 - val_accuracy: 0.6384
Epoch 9/30
152/152 [==============================] - 121s 793ms/step - loss: 2.3550 -
accuracy: 0.4824 - val_loss: 2.0676 - val_accuracy: 0.6391
Epoch 10/30
152/152 [==============================] - 120s 792ms/step - loss: 2.3414 -
accuracy: 0.4816 - val_loss: 2.0563 - val_accuracy: 0.6455
Epoch 11/30
152/152 [==============================] - 121s 793ms/step - loss: 2.3103 -
accuracy: 0.4931 - val_loss: 2.0467 - val_accuracy: 0.6462
Epoch 12/30
152/152 [==============================] - 120s 789ms/step - loss: 2.3317 -
accuracy: 0.4891 - val_loss: 2.0431 - val_accuracy: 0.6472
Epoch 13/30
152/152 [==============================] - 121s 793ms/step - loss: 2.2582 -
accuracy: 0.4953 - val_loss: 2.0373 - val_accuracy: 0.6430
Epoch 14/30
152/152 [==============================] - 120s 790ms/step - loss: 2.2749 -
accuracy: 0.4883 - val_loss: 2.0268 - val_accuracy: 0.6444
Epoch 15/30
152/152 [==============================] - 121s 793ms/step - loss: 2.2712 -
accuracy: 0.4885 - val_loss: 2.0150 - val_accuracy: 0.6490
```

```
Epoch 16/30
152/152 [==============================] - 120s 790ms/step - loss: 2.3015 -
accuracy: 0.4875 - val_loss: 2.0008 - val_accuracy: 0.6487
Epoch 17/30
152/152 [==============================] - 120s 791ms/step - loss: 2.2605 -
accuracy: 0.4936 - val_loss: 1.9936 - val_accuracy: 0.6480
Epoch 18/30
152/152 [==============================] - 120s 790ms/step - loss: 2.2367 -
accuracy: 0.5005 - val_loss: 1.9932 - val_accuracy: 0.6544
Epoch 19/30
152/152 [==============================] - 120s 791ms/step - loss: 2.2533 -
accuracy: 0.5012 - val_loss: 1.9928 - val_accuracy: 0.6497
Epoch 20/30
152/152 [==============================] - 120s 790ms/step - loss: 2.2389 -
accuracy: 0.4970 - val_loss: 1.9849 - val_accuracy: 0.6487
Epoch 21/30
152/152 [==============================] - 120s 791ms/step - loss: 2.2316 -
accuracy: 0.4960 - val_loss: 1.9630 - val_accuracy: 0.6519
Epoch 22/30
152/152 [==============================] - 120s 790ms/step - loss: 2.1946 -
accuracy: 0.5060 - val_loss: 1.9583 - val_accuracy: 0.6529
Epoch 23/30
152/152 [==============================] - 121s 794ms/step - loss: 2.2257 -
accuracy: 0.5035 - val_loss: 1.9436 - val_accuracy: 0.6568
Epoch 24/30
152/152 [==============================] - 121s 793ms/step - loss: 2.1634 -
accuracy: 0.5039 - val_loss: 1.9545 - val_accuracy: 0.6597
Epoch 25/30
152/152 [==============================] - 121s 794ms/step - loss: 2.1823 -
accuracy: 0.5069 - val_loss: 1.9456 - val_accuracy: 0.6572
Epoch 26/30
152/152 [==============================] - 120s 791ms/step - loss: 2.1758 -
accuracy: 0.4989 - val_loss: 1.9389 - val_accuracy: 0.6622
Epoch 27/30
152/152 [==============================] - 120s 791ms/step - loss: 2.1854 -
accuracy: 0.5002 - val_loss: 1.9336 - val_accuracy: 0.6568
Epoch 28/30
152/152 [==============================] - 121s 794ms/step - loss: 2.1693 -
accuracy: 0.5052 - val_loss: 1.9196 - val_accuracy: 0.6622
Epoch 29/30
152/152 [==============================] - 120s 792ms/step - loss: 2.1704 -
accuracy: 0.5046 - val_loss: 1.9136 - val_accuracy: 0.6636
Epoch 30/30
152/152 [==============================] - 120s 790ms/step - loss: 2.1367 -
accuracy: 0.5151 - val_loss: 1.9114 - val_accuracy: 0.6615
```

[46]:
```
test_loss, test_acc = model_mobilenet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 8s 59ms/step - loss: 1.9170 -
accuracy: 0.6685
Test Accuracy: 0.6685
```

[47]:
```
plt.plot(history_mobilenet2.history['loss'], label='Training Loss')
plt.plot(history_mobilenet2.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




# 7 MobileNet3 learning rate=0.0001 Nadam

[74]:
```
optimizer = tf.keras.optimizers.Nadam(
    learning_rate=0.0001,
    beta_1=0.9,  #
    beta_2=0.999  #
```

```
)
model_resnet.compile(
    optimizer=optimizer,
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

early_stop = EarlyStopping(monitor='val_accuracy', patience=5,␣
 ↪restore_best_weights=True)
```

[76]:
```
history_mobilenet3 = model_mobilenet.fit(
    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict,
)
```

```
Epoch 1/30
152/152 [==============================] - 120s 790ms/step - loss: 2.0564 -
accuracy: 0.5292 - val_loss: 1.8656 - val_accuracy: 0.6728
Epoch 2/30
152/152 [==============================] - 120s 788ms/step - loss: 2.0962 -
accuracy: 0.5148 - val_loss: 1.8630 - val_accuracy: 0.6710
Epoch 3/30
152/152 [==============================] - 120s 786ms/step - loss: 2.0801 -
accuracy: 0.5229 - val_loss: 1.8627 - val_accuracy: 0.6714
Epoch 4/30
152/152 [==============================] - 120s 788ms/step - loss: 2.0850 -
accuracy: 0.5152 - val_loss: 1.8621 - val_accuracy: 0.6735
Epoch 5/30
152/152 [==============================] - 120s 790ms/step - loss: 2.0637 -
accuracy: 0.5174 - val_loss: 1.8617 - val_accuracy: 0.6732
Epoch 6/30
152/152 [==============================] - 120s 788ms/step - loss: 2.0819 -
accuracy: 0.5217 - val_loss: 1.8625 - val_accuracy: 0.6693
Epoch 7/30
152/152 [==============================] - 120s 787ms/step - loss: 2.0758 -
accuracy: 0.5172 - val_loss: 1.8599 - val_accuracy: 0.6700
Epoch 8/30
152/152 [==============================] - 119s 785ms/step - loss: 2.0807 -
accuracy: 0.5244 - val_loss: 1.8590 - val_accuracy: 0.6718
Epoch 9/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0741 -
accuracy: 0.5203 - val_loss: 1.8578 - val_accuracy: 0.6732
Epoch 10/30
152/152 [==============================] - 119s 786ms/step - loss: 2.0670 -
accuracy: 0.5211 - val_loss: 1.8588 - val_accuracy: 0.6718
Epoch 11/30
```

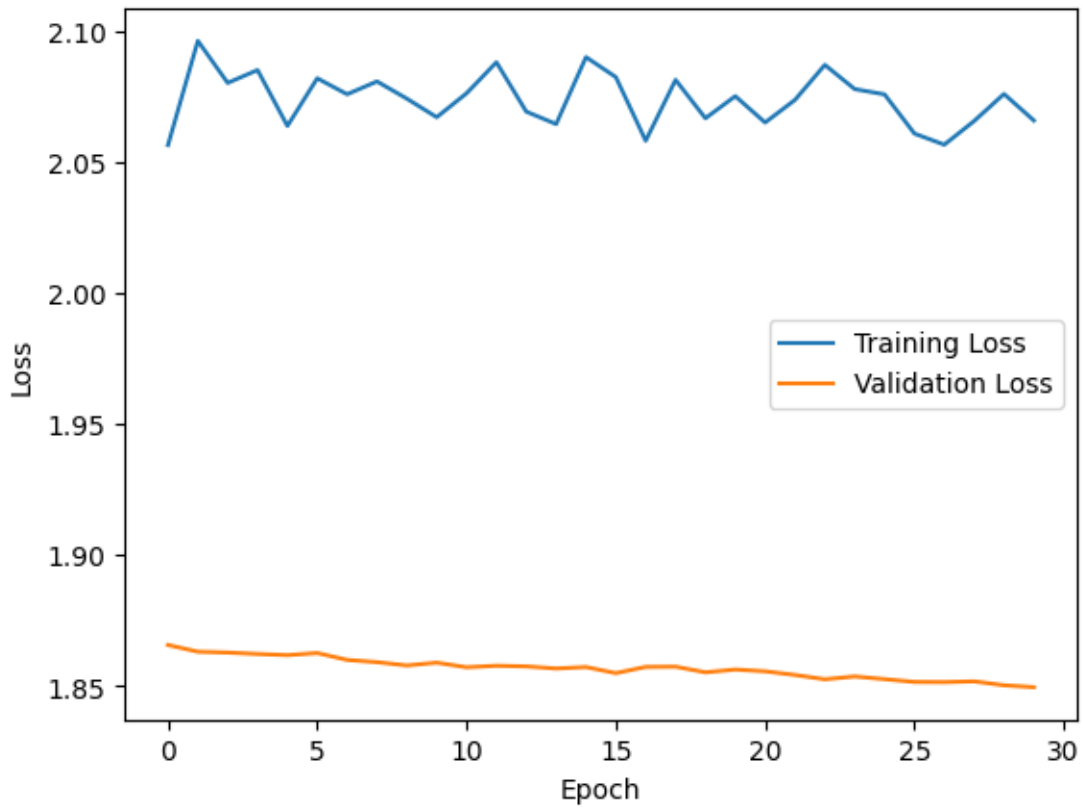
```
152/152 [==============================] - 120s 788ms/step - loss: 2.0761 -
accuracy: 0.5209 - val_loss: 1.8571 - val_accuracy: 0.6710
Epoch 12/30
152/152 [==============================] - 120s 787ms/step - loss: 2.0880 -
accuracy: 0.5181 - val_loss: 1.8576 - val_accuracy: 0.6718
Epoch 13/30
152/152 [==============================] - 119s 785ms/step - loss: 2.0692 -
accuracy: 0.5195 - val_loss: 1.8574 - val_accuracy: 0.6721
Epoch 14/30
152/152 [==============================] - 120s 787ms/step - loss: 2.0644 -
accuracy: 0.5198 - val_loss: 1.8566 - val_accuracy: 0.6721
Epoch 15/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0900 -
accuracy: 0.5193 - val_loss: 1.8571 - val_accuracy: 0.6725
Epoch 16/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0823 -
accuracy: 0.5135 - val_loss: 1.8548 - val_accuracy: 0.6728
Epoch 17/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0580 -
accuracy: 0.5206 - val_loss: 1.8572 - val_accuracy: 0.6739
Epoch 18/30
152/152 [==============================] - 120s 787ms/step - loss: 2.0813 -
accuracy: 0.5181 - val_loss: 1.8573 - val_accuracy: 0.6725
Epoch 19/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0666 -
accuracy: 0.5239 - val_loss: 1.8551 - val_accuracy: 0.6714
Epoch 20/30
152/152 [==============================] - 120s 788ms/step - loss: 2.0751 -
accuracy: 0.5223 - val_loss: 1.8562 - val_accuracy: 0.6696
Epoch 21/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0650 -
accuracy: 0.5248 - val_loss: 1.8555 - val_accuracy: 0.6714
Epoch 22/30
152/152 [==============================] - 120s 790ms/step - loss: 2.0736 -
accuracy: 0.5165 - val_loss: 1.8541 - val_accuracy: 0.6714
Epoch 23/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0870 -
accuracy: 0.5146 - val_loss: 1.8525 - val_accuracy: 0.6725
Epoch 24/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0778 -
accuracy: 0.5212 - val_loss: 1.8536 - val_accuracy: 0.6721
Epoch 25/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0757 -
accuracy: 0.5209 - val_loss: 1.8525 - val_accuracy: 0.6721
Epoch 26/30
152/152 [==============================] - 120s 792ms/step - loss: 2.0608 -
accuracy: 0.5247 - val_loss: 1.8515 - val_accuracy: 0.6718
Epoch 27/30
```

```
152/152 [==============================] - 121s 794ms/step - loss: 2.0565 -
accuracy: 0.5184 - val_loss: 1.8514 - val_accuracy: 0.6735
Epoch 28/30
152/152 [==============================] - 120s 789ms/step - loss: 2.0656 -
accuracy: 0.5214 - val_loss: 1.8517 - val_accuracy: 0.6725
Epoch 29/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0759 -
accuracy: 0.5233 - val_loss: 1.8502 - val_accuracy: 0.6725
Epoch 30/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0658 -
accuracy: 0.5239 - val_loss: 1.8495 - val_accuracy: 0.6707
```

[77]:
```
test_loss, test_acc = model_mobilenet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 8s 60ms/step - loss: 1.8482 -
accuracy: 0.6834
Test Accuracy: 0.6834
```

[78]:
```
plt.plot(history_mobilenet3.history['loss'], label='Training Loss')
plt.plot(history_mobilenet3.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




## 8 MobileNet4 learning rate=0.0001 SGD

[80]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.0001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = tf.keras.optimizers.SGD(learning_rate=lr_schedule, momentum=0.9)
model_mobilenet.compile(optimizer=optimizer, loss='categorical_crossentropy',␣
 ↪metrics=['accuracy'])

early_stop = EarlyStopping(monitor='val_accuracy', patience=5,␣
 ↪restore_best_weights=True)
```

[81]:
```
history_mobilenet4 = model_mobilenet.fit(
    train_dataset,
    epochs=30,
    validation_data=val_dataset,
    class_weight=class_weight_dict,
)
```

```
Epoch 1/30
```

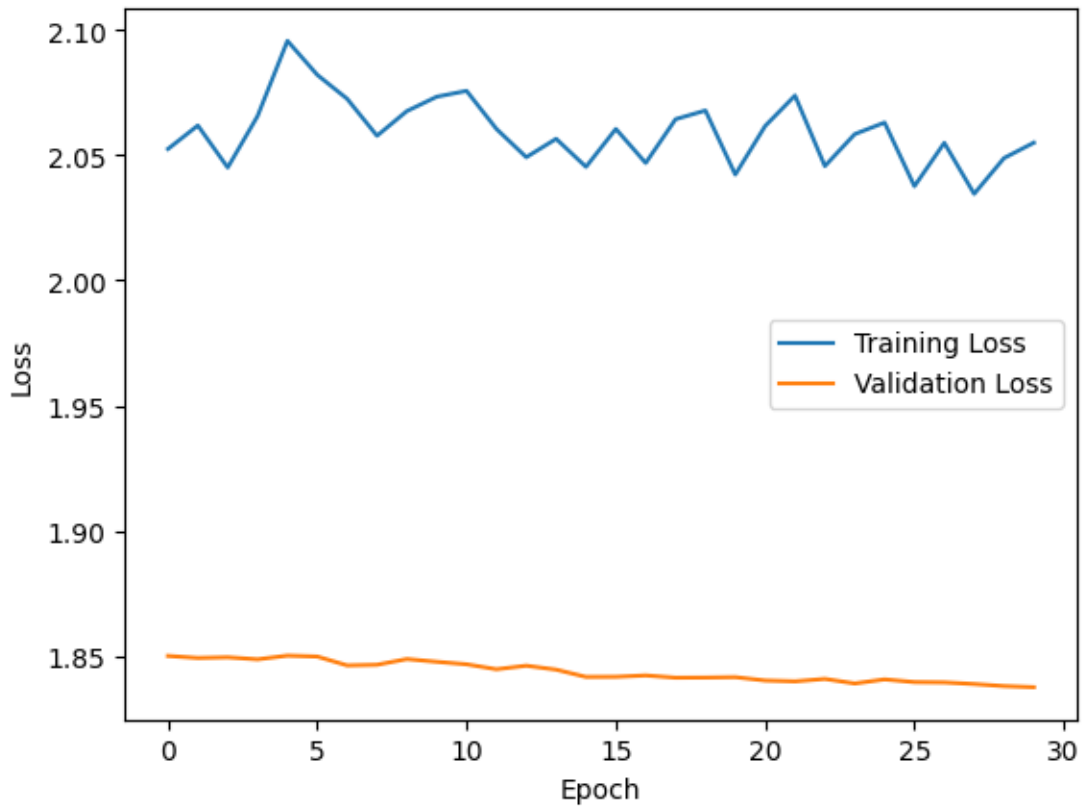
```
152/152 [==============================] - 122s 791ms/step - loss: 2.0527 -
accuracy: 0.5268 - val_loss: 1.8503 - val_accuracy: 0.6714
Epoch 2/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0824 -
accuracy: 0.5122 - val_loss: 1.8495 - val_accuracy: 0.6721
Epoch 3/30
152/152 [==============================] - 121s 795ms/step - loss: 2.0483 -
accuracy: 0.5290 - val_loss: 1.8497 - val_accuracy: 0.6725
Epoch 4/30
152/152 [==============================] - 121s 796ms/step - loss: 2.0723 -
accuracy: 0.5219 - val_loss: 1.8489 - val_accuracy: 0.6725
Epoch 5/30
152/152 [==============================] - 121s 795ms/step - loss: 2.1189 -
accuracy: 0.5139 - val_loss: 1.8504 - val_accuracy: 0.6710
Epoch 6/30
152/152 [==============================] - 120s 792ms/step - loss: 2.0921 -
accuracy: 0.5135 - val_loss: 1.8500 - val_accuracy: 0.6718
Epoch 7/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0844 -
accuracy: 0.5206 - val_loss: 1.8465 - val_accuracy: 0.6725
Epoch 8/30
152/152 [==============================] - 120s 790ms/step - loss: 2.0558 -
accuracy: 0.5213 - val_loss: 1.8468 - val_accuracy: 0.6721
Epoch 9/30
152/152 [==============================] - 120s 792ms/step - loss: 2.0440 -
accuracy: 0.5260 - val_loss: 1.8490 - val_accuracy: 0.6721
Epoch 10/30
152/152 [==============================] - 120s 792ms/step - loss: 2.0662 -
accuracy: 0.5198 - val_loss: 1.8479 - val_accuracy: 0.6714
Epoch 11/30
152/152 [==============================] - 121s 794ms/step - loss: 2.0588 -
accuracy: 0.5226 - val_loss: 1.8469 - val_accuracy: 0.6739
Epoch 12/30
152/152 [==============================] - 120s 790ms/step - loss: 2.0581 -
accuracy: 0.5221 - val_loss: 1.8450 - val_accuracy: 0.6728
Epoch 13/30
152/152 [==============================] - 120s 792ms/step - loss: 2.0512 -
accuracy: 0.5262 - val_loss: 1.8464 - val_accuracy: 0.6746
Epoch 14/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0612 -
accuracy: 0.5212 - val_loss: 1.8448 - val_accuracy: 0.6750
Epoch 15/30
152/152 [==============================] - 121s 795ms/step - loss: 2.0309 -
accuracy: 0.5270 - val_loss: 1.8419 - val_accuracy: 0.6746
Epoch 16/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0645 -
accuracy: 0.5202 - val_loss: 1.8420 - val_accuracy: 0.6742
Epoch 17/30
```

```
152/152 [==============================] - 120s 791ms/step - loss: 2.0317 -
accuracy: 0.5275 - val_loss: 1.8425 - val_accuracy: 0.6757
Epoch 18/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0490 -
accuracy: 0.5198 - val_loss: 1.8416 - val_accuracy: 0.6732
Epoch 19/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0674 -
accuracy: 0.5149 - val_loss: 1.8417 - val_accuracy: 0.6739
Epoch 20/30
152/152 [==============================] - 121s 797ms/step - loss: 2.0176 -
accuracy: 0.5334 - val_loss: 1.8418 - val_accuracy: 0.6750
Epoch 21/30
152/152 [==============================] - 120s 791ms/step - loss: 2.0783 -
accuracy: 0.5248 - val_loss: 1.8405 - val_accuracy: 0.6771
Epoch 22/30
152/152 [==============================] - 121s 792ms/step - loss: 2.0792 -
accuracy: 0.5269 - val_loss: 1.8401 - val_accuracy: 0.6757
Epoch 23/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0815 -
accuracy: 0.5150 - val_loss: 1.8411 - val_accuracy: 0.6760
Epoch 24/30
152/152 [==============================] - 120s 792ms/step - loss: 2.0386 -
accuracy: 0.5274 - val_loss: 1.8393 - val_accuracy: 0.6750
Epoch 25/30
152/152 [==============================] - 121s 794ms/step - loss: 2.0418 -
accuracy: 0.5269 - val_loss: 1.8409 - val_accuracy: 0.6753
Epoch 26/30
152/152 [==============================] - 121s 792ms/step - loss: 2.0358 -
accuracy: 0.5301 - val_loss: 1.8398 - val_accuracy: 0.6753
Epoch 27/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0266 -
accuracy: 0.5301 - val_loss: 1.8397 - val_accuracy: 0.6753
Epoch 28/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0209 -
accuracy: 0.5296 - val_loss: 1.8391 - val_accuracy: 0.6757
Epoch 29/30
152/152 [==============================] - 121s 793ms/step - loss: 2.0468 -
accuracy: 0.5197 - val_loss: 1.8383 - val_accuracy: 0.6764
Epoch 30/30
152/152 [==============================] - 120s 793ms/step - loss: 2.0949 -
accuracy: 0.5173 - val_loss: 1.8378 - val_accuracy: 0.6746
```

[82]:
```
test_loss, test_acc = model_mobilenet.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 8s 59ms/step - loss: 1.8375 -
accuracy: 0.6858
Test Accuracy: 0.6858
```

[83]:
```
plt.plot(history_mobilenet4.history['loss'], label='Training Loss')
plt.plot(history_mobilenet4.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




# 9 CNN1 Learning Rate = 0.001 Adam

[7]:
```
def pre_process_and_build_image_dataset(folder_path):
    for root, dirs, files in os.walk(folder_path):
        for poke in dirs:
            poke_dir = os.path.join(root, poke)
            for image in os.listdir(poke_dir):
                original_image_path = os.path.join(poke_dir, image)

                #     BGR
                img = cv2.imread(original_image_path)
                if img is None:
```

```
                    print(f"Warning: Failed to read image␣
↪{original_image_path}")
                    continue

                #    32x32
                img_resized = cv2.resize(img, (32, 32))
                cv2.imwrite(original_image_path, img_resized)  #

    print(f"Preprocessing completed for {folder_path}")
```

[9]:
```
pre_process_and_build_image_dataset(train_path)
pre_process_and_build_image_dataset(test_path)
pre_process_and_build_image_dataset(val_path)
```

```
Preprocessing completed for D:/files/LUND/DeepLearning/project/pokemon-
dataset-1000/train
Preprocessing completed for D:/files/LUND/DeepLearning/project/pokemon-
dataset-1000/test
Preprocessing completed for D:/files/LUND/DeepLearning/project/pokemon-
dataset-1000/val
```

[11]:
```
train_dataset = train_datagen.flow_from_directory(
    train_path,
    target_size=(32, 32),  #
    batch_size=128,
    class_mode='categorical',  #
    seed=42
)
```

```
Found 19434 images belonging to 1000 classes.
```

[12]:
```
val_dataset = image_dataset_from_directory(
    val_path,
    label_mode="categorical",
    image_size=(32, 32),
    seed=42
)
```

```
Found 2815 files belonging to 1000 classes.
```

[13]:
```
test_dataset = image_dataset_from_directory(
    test_path,
    label_mode="categorical",
    image_size=(32, 32),
    seed=42
)
```

```
Found 4290 files belonging to 1000 classes.
```

[15]:
```
model_CNN = models.Sequential([
    layers.Input(shape=(32, 32, 3)),

    layers.Conv2D(filters=32, kernel_size=(3, 3), activation='relu'),
    layers.MaxPool2D(pool_size=(2, 2)),
    layers.Conv2D(filters=64, kernel_size=(3, 3), activation='relu'),
    layers.MaxPool2D(pool_size=(2, 2)),
    layers.Conv2D(filters=128, kernel_size=(3, 3), activation='relu'),
    layers.Conv2D(filters=256, kernel_size=(3, 3), activation='relu'),
    layers.Flatten(),  #
    layers.Dense(256, activation="relu"),
    layers.Dense(512, activation="relu"),
    layers.Dense(1024, activation="relu"),
    layers.Dense(2048, activation="relu"),
    layers.Dense(1000, activation="softmax")
])
```

[19]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = Adam(learning_rate=lr_schedule)

model_CNN.compile(
    optimizer=optimizer,
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

[21]:
```
history_CNN1 = model_CNN.fit(
    train_dataset,
    epochs=30,
    batch_size=128,
    validation_data=val_dataset
)
```

```
Epoch 1/30
152/152 [==============================] - 163s 1s/step - loss: 6.8625 -
accuracy: 0.0011 - val_loss: 6.3684 - val_accuracy: 0.0046
Epoch 2/30
152/152 [==============================] - 13s 84ms/step - loss: 6.1781 -
accuracy: 0.0066 - val_loss: 6.1341 - val_accuracy: 0.0075
Epoch 3/30
152/152 [==============================] - 13s 85ms/step - loss: 5.9670 -
accuracy: 0.0083 - val_loss: 6.0156 - val_accuracy: 0.0099
Epoch 4/30
152/152 [==============================] - 13s 84ms/step - loss: 5.6852 -
accuracy: 0.0163 - val_loss: 6.2549 - val_accuracy: 0.0171
```

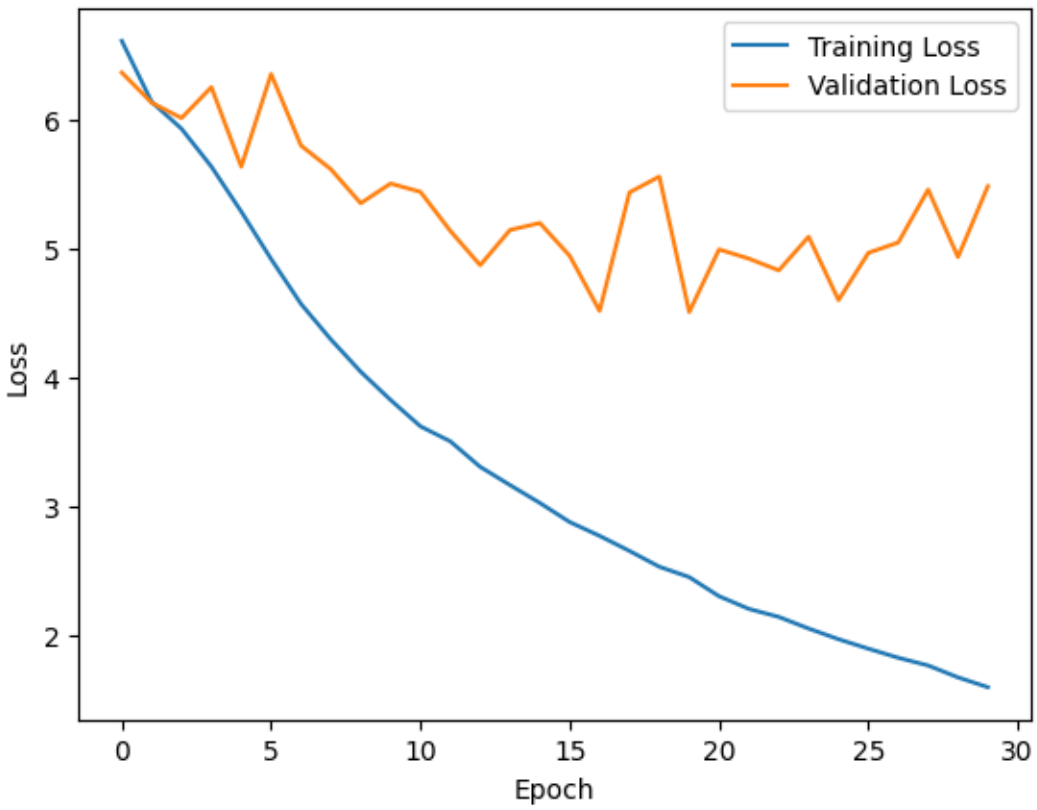
```
Epoch 5/30
152/152 [==============================] - 13s 85ms/step - loss: 5.3527 -
accuracy: 0.0283 - val_loss: 5.6383 - val_accuracy: 0.0306
Epoch 6/30
152/152 [==============================] - 13s 84ms/step - loss: 4.9516 -
accuracy: 0.0502 - val_loss: 6.3571 - val_accuracy: 0.0249
Epoch 7/30
152/152 [==============================] - 13s 84ms/step - loss: 4.5899 -
accuracy: 0.0706 - val_loss: 5.8021 - val_accuracy: 0.0412
Epoch 8/30
152/152 [==============================] - 13s 85ms/step - loss: 4.3006 -
accuracy: 0.0979 - val_loss: 5.6186 - val_accuracy: 0.0561
Epoch 9/30
152/152 [==============================] - 13s 83ms/step - loss: 4.0287 -
accuracy: 0.1257 - val_loss: 5.3543 - val_accuracy: 0.0750
Epoch 10/30
152/152 [==============================] - 13s 84ms/step - loss: 3.8071 -
accuracy: 0.1445 - val_loss: 5.5073 - val_accuracy: 0.0686
Epoch 11/30
152/152 [==============================] - 13s 84ms/step - loss: 3.6179 -
accuracy: 0.1708 - val_loss: 5.4429 - val_accuracy: 0.0881
Epoch 12/30
152/152 [==============================] - 13s 86ms/step - loss: 3.4352 -
accuracy: 0.1948 - val_loss: 5.1404 - val_accuracy: 0.0959
Epoch 13/30
152/152 [==============================] - 13s 84ms/step - loss: 3.2557 -
accuracy: 0.2203 - val_loss: 4.8737 - val_accuracy: 0.1275
Epoch 14/30
152/152 [==============================] - 13s 85ms/step - loss: 3.1431 -
accuracy: 0.2355 - val_loss: 5.1470 - val_accuracy: 0.1350
Epoch 15/30
152/152 [==============================] - 13s 86ms/step - loss: 2.9855 -
accuracy: 0.2605 - val_loss: 5.2009 - val_accuracy: 0.1300
Epoch 16/30
152/152 [==============================] - 13s 84ms/step - loss: 2.8386 -
accuracy: 0.2900 - val_loss: 4.9458 - val_accuracy: 0.1517
Epoch 17/30
152/152 [==============================] - 13s 85ms/step - loss: 2.7267 -
accuracy: 0.3062 - val_loss: 4.5188 - val_accuracy: 0.1783
Epoch 18/30
152/152 [==============================] - 13s 84ms/step - loss: 2.6156 -
accuracy: 0.3333 - val_loss: 5.4384 - val_accuracy: 0.1393
Epoch 19/30
152/152 [==============================] - 13s 84ms/step - loss: 2.5001 -
accuracy: 0.3533 - val_loss: 5.5614 - val_accuracy: 0.1389
Epoch 20/30
152/152 [==============================] - 13s 84ms/step - loss: 2.4236 -
accuracy: 0.3723 - val_loss: 4.5100 - val_accuracy: 0.2000
```

```
Epoch 21/30
152/152 [==============================] - 13s 84ms/step - loss: 2.2454 -
accuracy: 0.3988 - val_loss: 4.9961 - val_accuracy: 0.1822
Epoch 22/30
152/152 [==============================] - 13s 84ms/step - loss: 2.1620 -
accuracy: 0.4261 - val_loss: 4.9244 - val_accuracy: 0.1890
Epoch 23/30
152/152 [==============================] - 13s 84ms/step - loss: 2.0838 -
accuracy: 0.4368 - val_loss: 4.8334 - val_accuracy: 0.2067
Epoch 24/30
152/152 [==============================] - 13s 86ms/step - loss: 1.9889 -
accuracy: 0.4593 - val_loss: 5.0947 - val_accuracy: 0.1982
Epoch 25/30
152/152 [==============================] - 13s 83ms/step - loss: 1.9179 -
accuracy: 0.4721 - val_loss: 4.6027 - val_accuracy: 0.2313
Epoch 26/30
152/152 [==============================] - 13s 85ms/step - loss: 1.8782 -
accuracy: 0.4817 - val_loss: 4.9694 - val_accuracy: 0.2188
Epoch 27/30
152/152 [==============================] - 13s 84ms/step - loss: 1.8007 -
accuracy: 0.5049 - val_loss: 5.0497 - val_accuracy: 0.2284
Epoch 28/30
152/152 [==============================] - 13s 84ms/step - loss: 1.7278 -
accuracy: 0.5182 - val_loss: 5.4604 - val_accuracy: 0.2178
Epoch 29/30
152/152 [==============================] - 13s 84ms/step - loss: 1.5952 -
accuracy: 0.5470 - val_loss: 4.9372 - val_accuracy: 0.2448
Epoch 30/30
152/152 [==============================] - 13s 84ms/step - loss: 1.5391 -
accuracy: 0.5665 - val_loss: 5.4863 - val_accuracy: 0.2185
```

[22]:
```
test_loss, test_acc = model_CNN.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 30s 212ms/step - loss: 5.4619 -
accuracy: 0.2212
Test Accuracy: 0.2212
```

[23]:
```
plt.plot(history_CNN1.history['loss'], label='Training Loss')
plt.plot(history_CNN1.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




## 10 CNN Learning Rate = 0.0001 Adam

[25]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.0001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = Adam(learning_rate=lr_schedule)

model_CNN.compile(
    optimizer=optimizer,
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

[26]:
```
history_CNN2 = model_CNN.fit(
    train_dataset,
    epochs=30,
    batch_size=128,
    validation_data=val_dataset
)
```

```
Epoch 1/30
152/152 [==============================] - 13s 85ms/step - loss: 1.2373 -
accuracy: 0.6456 - val_loss: 5.4595 - val_accuracy: 0.2607
Epoch 2/30
152/152 [==============================] - 13s 82ms/step - loss: 1.0388 -
accuracy: 0.7033 - val_loss: 5.8750 - val_accuracy: 0.2679
Epoch 3/30
152/152 [==============================] - 13s 83ms/step - loss: 0.9542 -
accuracy: 0.7251 - val_loss: 5.6130 - val_accuracy: 0.2799
Epoch 4/30
152/152 [==============================] - 13s 82ms/step - loss: 0.9563 -
accuracy: 0.7278 - val_loss: 6.1015 - val_accuracy: 0.2600
Epoch 5/30
152/152 [==============================] - 13s 83ms/step - loss: 0.9053 -
accuracy: 0.7430 - val_loss: 5.8844 - val_accuracy: 0.2853
Epoch 6/30
152/152 [==============================] - 13s 83ms/step - loss: 0.8805 -
accuracy: 0.7455 - val_loss: 5.6100 - val_accuracy: 0.3076
Epoch 7/30
152/152 [==============================] - 13s 83ms/step - loss: 0.8787 -
accuracy: 0.7457 - val_loss: 6.2501 - val_accuracy: 0.2831
Epoch 8/30
152/152 [==============================] - 13s 84ms/step - loss: 0.8083 -
accuracy: 0.7606 - val_loss: 6.2532 - val_accuracy: 0.2792
Epoch 9/30
152/152 [==============================] - 13s 82ms/step - loss: 0.8255 -
accuracy: 0.7575 - val_loss: 5.9057 - val_accuracy: 0.3002
Epoch 10/30
152/152 [==============================] - 13s 83ms/step - loss: 0.7863 -
accuracy: 0.7759 - val_loss: 6.5839 - val_accuracy: 0.2689
Epoch 11/30
152/152 [==============================] - 13s 84ms/step - loss: 0.7855 -
accuracy: 0.7693 - val_loss: 6.4029 - val_accuracy: 0.2817
Epoch 12/30
152/152 [==============================] - 13s 82ms/step - loss: 0.7842 -
accuracy: 0.7705 - val_loss: 6.2813 - val_accuracy: 0.2973
Epoch 13/30
152/152 [==============================] - 13s 83ms/step - loss: 0.7713 -
accuracy: 0.7723 - val_loss: 6.5152 - val_accuracy: 0.2892
Epoch 14/30
152/152 [==============================] - 13s 82ms/step - loss: 0.7594 -
accuracy: 0.7796 - val_loss: 6.4421 - val_accuracy: 0.2998
Epoch 15/30
152/152 [==============================] - 13s 83ms/step - loss: 0.7636 -
accuracy: 0.7823 - val_loss: 6.4081 - val_accuracy: 0.2963
Epoch 16/30
152/152 [==============================] - 13s 83ms/step - loss: 0.7213 -
accuracy: 0.7914 - val_loss: 6.2535 - val_accuracy: 0.3080
```

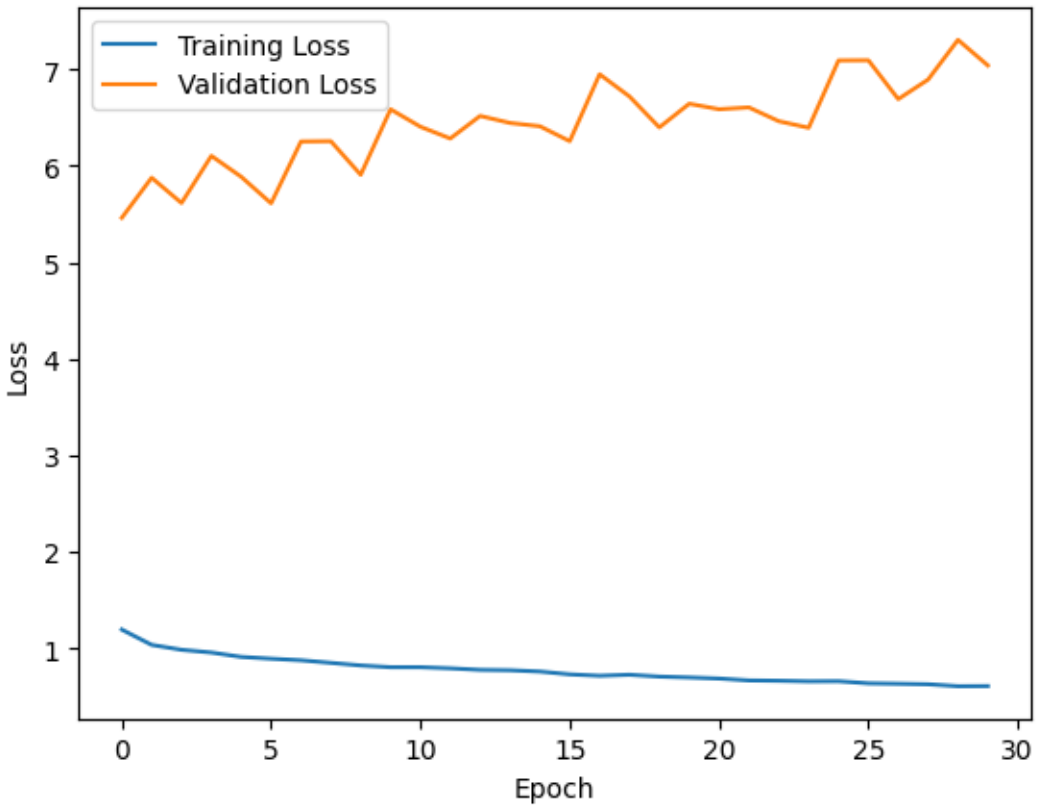
```
Epoch 17/30
152/152 [==============================] - 13s 83ms/step - loss: 0.7091 -
accuracy: 0.7948 - val_loss: 6.9467 - val_accuracy: 0.2792
Epoch 18/30
152/152 [==============================] - 13s 83ms/step - loss: 0.7253 -
accuracy: 0.7910 - val_loss: 6.7165 - val_accuracy: 0.2892
Epoch 19/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6919 -
accuracy: 0.7998 - val_loss: 6.3962 - val_accuracy: 0.3115
Epoch 20/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6739 -
accuracy: 0.8040 - val_loss: 6.6413 - val_accuracy: 0.3023
Epoch 21/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6848 -
accuracy: 0.8049 - val_loss: 6.5843 - val_accuracy: 0.3059
Epoch 22/30
152/152 [==============================] - 12s 82ms/step - loss: 0.6526 -
accuracy: 0.8084 - val_loss: 6.6031 - val_accuracy: 0.3101
Epoch 23/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6473 -
accuracy: 0.8095 - val_loss: 6.4602 - val_accuracy: 0.3208
Epoch 24/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6427 -
accuracy: 0.8101 - val_loss: 6.3922 - val_accuracy: 0.3261
Epoch 25/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6414 -
accuracy: 0.8135 - val_loss: 7.0899 - val_accuracy: 0.2899
Epoch 26/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6274 -
accuracy: 0.8175 - val_loss: 7.0916 - val_accuracy: 0.3005
Epoch 27/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6137 -
accuracy: 0.8206 - val_loss: 6.6892 - val_accuracy: 0.3062
Epoch 28/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6144 -
accuracy: 0.8220 - val_loss: 6.8919 - val_accuracy: 0.3055
Epoch 29/30
152/152 [==============================] - 13s 83ms/step - loss: 0.6154 -
accuracy: 0.8224 - val_loss: 7.3030 - val_accuracy: 0.2849
Epoch 30/30
152/152 [==============================] - 13s 84ms/step - loss: 0.5895 -
accuracy: 0.8286 - val_loss: 7.0380 - val_accuracy: 0.3020
```

[27]:
```
test_loss, test_acc = model_CNN.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 3s 19ms/step - loss: 7.1933 -
accuracy: 0.3028
```

```
Test Accuracy: 0.3028
```

[28]:
```
plt.plot(history_CNN2.history['loss'], label='Training Loss')
plt.plot(history_CNN2.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```




## 11 CNN Learning Rate = 0.0001 Nadam

[30]:
```
optimizer = tf.keras.optimizers.Nadam(
    learning_rate=0.0001,
    beta_1=0.9,  #
    beta_2=0.999  #
)
model_CNN.compile(
    optimizer=optimizer,
    loss='categorical_crossentropy',
    metrics=['accuracy']
```

```
)
```

[31]:
```
history_CNN3 = model_CNN.fit(
    train_dataset,
    epochs=30,
    batch_size=128,
    validation_data=val_dataset
)
```

```
Epoch 1/30
152/152 [==============================] - 14s 84ms/step - loss: 0.5881 -
accuracy: 0.8278 - val_loss: 6.7580 - val_accuracy: 0.3222
Epoch 2/30
152/152 [==============================] - 13s 83ms/step - loss: 0.5935 -
accuracy: 0.8237 - val_loss: 6.7719 - val_accuracy: 0.3268
Epoch 3/30
152/152 [==============================] - 13s 83ms/step - loss: 0.5908 -
accuracy: 0.8262 - val_loss: 7.4158 - val_accuracy: 0.2792
Epoch 4/30
152/152 [==============================] - 13s 84ms/step - loss: 0.5719 -
accuracy: 0.8309 - val_loss: 7.0426 - val_accuracy: 0.3215
Epoch 5/30
152/152 [==============================] - 13s 83ms/step - loss: 0.5827 -
accuracy: 0.8305 - val_loss: 6.6764 - val_accuracy: 0.3300
Epoch 6/30
152/152 [==============================] - 12s 77ms/step - loss: 0.5952 -
accuracy: 0.8230 - val_loss: 6.6991 - val_accuracy: 0.3211
Epoch 7/30
152/152 [==============================] - 12s 77ms/step - loss: 0.5522 -
accuracy: 0.8413 - val_loss: 6.8053 - val_accuracy: 0.3222
Epoch 8/30
152/152 [==============================] - 13s 83ms/step - loss: 0.5378 -
accuracy: 0.8449 - val_loss: 6.8515 - val_accuracy: 0.3258
Epoch 9/30
152/152 [==============================] - 11s 75ms/step - loss: 0.5324 -
accuracy: 0.8449 - val_loss: 7.0057 - val_accuracy: 0.3194
Epoch 10/30
152/152 [==============================] - 12s 76ms/step - loss: 0.5293 -
accuracy: 0.8482 - val_loss: 7.3378 - val_accuracy: 0.3101
Epoch 11/30
152/152 [==============================] - 12s 76ms/step - loss: 0.5396 -
accuracy: 0.8420 - val_loss: 7.3908 - val_accuracy: 0.2995
Epoch 12/30
152/152 [==============================] - 12s 76ms/step - loss: 0.5289 -
accuracy: 0.8423 - val_loss: 7.1534 - val_accuracy: 0.3140
Epoch 13/30
152/152 [==============================] - 11s 75ms/step - loss: 0.5293 -
accuracy: 0.8456 - val_loss: 6.5561 - val_accuracy: 0.3375
```

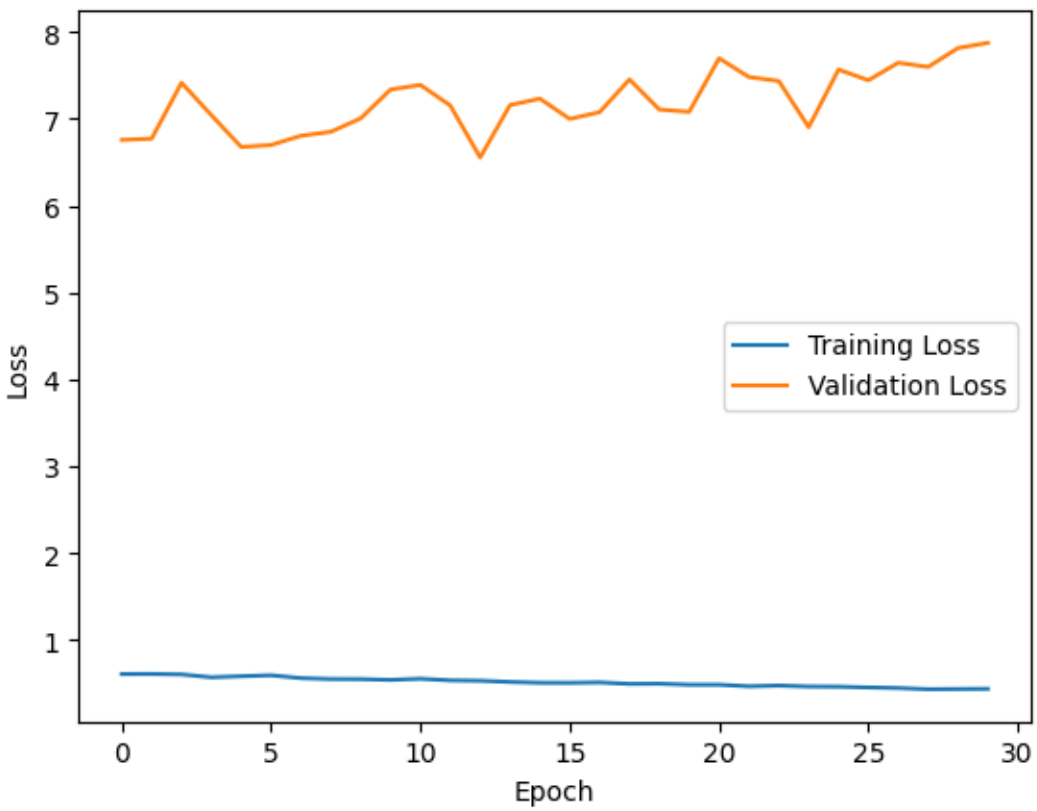
```
Epoch 14/30
152/152 [==============================] - 12s 79ms/step - loss: 0.5134 -
accuracy: 0.8490 - val_loss: 7.1588 - val_accuracy: 0.3201
Epoch 15/30
152/152 [==============================] - 13s 84ms/step - loss: 0.5000 -
accuracy: 0.8544 - val_loss: 7.2327 - val_accuracy: 0.3208
Epoch 16/30
152/152 [==============================] - 13s 83ms/step - loss: 0.5250 -
accuracy: 0.8443 - val_loss: 6.9993 - val_accuracy: 0.3329
Epoch 17/30
152/152 [==============================] - 12s 78ms/step - loss: 0.4894 -
accuracy: 0.8561 - val_loss: 7.0775 - val_accuracy: 0.3218
Epoch 18/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4833 -
accuracy: 0.8571 - val_loss: 7.4541 - val_accuracy: 0.3172
Epoch 19/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4795 -
accuracy: 0.8608 - val_loss: 7.1070 - val_accuracy: 0.3226
Epoch 20/30
152/152 [==============================] - 11s 75ms/step - loss: 0.4821 -
accuracy: 0.8583 - val_loss: 7.0812 - val_accuracy: 0.3290
Epoch 21/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4664 -
accuracy: 0.8611 - val_loss: 7.6977 - val_accuracy: 0.3105
Epoch 22/30
152/152 [==============================] - 14s 93ms/step - loss: 0.4574 -
accuracy: 0.8649 - val_loss: 7.4817 - val_accuracy: 0.3133
Epoch 23/30
152/152 [==============================] - 14s 89ms/step - loss: 0.4802 -
accuracy: 0.8661 - val_loss: 7.4357 - val_accuracy: 0.3158
Epoch 24/30
152/152 [==============================] - 17s 109ms/step - loss: 0.4662 -
accuracy: 0.8618 - val_loss: 6.9044 - val_accuracy: 0.3556
Epoch 25/30
152/152 [==============================] - 113s 750ms/step - loss: 0.4648 -
accuracy: 0.8647 - val_loss: 7.5676 - val_accuracy: 0.3240
Epoch 26/30
152/152 [==============================] - 100s 656ms/step - loss: 0.4423 -
accuracy: 0.8734 - val_loss: 7.4444 - val_accuracy: 0.3321
Epoch 27/30
152/152 [==============================] - 11s 75ms/step - loss: 0.4257 -
accuracy: 0.8742 - val_loss: 7.6465 - val_accuracy: 0.3151
Epoch 28/30
152/152 [==============================] - 11s 75ms/step - loss: 0.4267 -
accuracy: 0.8696 - val_loss: 7.5977 - val_accuracy: 0.3243
Epoch 29/30
152/152 [==============================] - 12s 78ms/step - loss: 0.4379 -
accuracy: 0.8695 - val_loss: 7.8153 - val_accuracy: 0.3240
```

```
Epoch 30/30
152/152 [==============================] - 12s 79ms/step - loss: 0.4185 -
accuracy: 0.8745 - val_loss: 7.8751 - val_accuracy: 0.3162
```

[32]:
```
test_loss, test_acc = model_CNN.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 26s 179ms/step - loss: 8.1593 -
accuracy: 0.3166
Test Accuracy: 0.3166
```

[33]:
```
plt.plot(history_CNN3.history['loss'], label='Training Loss')
plt.plot(history_CNN3.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```



## 12 CNN4 Learning Rate = 0.0001 SGD

[35]:
```
lr_schedule = ExponentialDecay(initial_learning_rate=0.0001,
                               decay_steps=1000,
                               decay_rate=0.96,
                               staircase=True)
optimizer = tf.keras.optimizers.SGD(learning_rate=lr_schedule, momentum=0.9)
model_CNN.compile(optimizer=optimizer, loss='categorical_crossentropy',␣
 ↪metrics=['accuracy'])
```

[36]:
```
history_CNN4 = model_CNN.fit(
    train_dataset,
    epochs=30,
    batch_size=128,
    validation_data=val_dataset
)
```

```
Epoch 1/30
152/152 [==============================] - 12s 77ms/step - loss: 0.8333 -
accuracy: 0.7933 - val_loss: 7.9229 - val_accuracy: 0.2948
Epoch 2/30
152/152 [==============================] - 12s 78ms/step - loss: 0.4899 -
accuracy: 0.8549 - val_loss: 7.3378 - val_accuracy: 0.3488
Epoch 3/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4739 -
accuracy: 0.8611 - val_loss: 8.1903 - val_accuracy: 0.3023
Epoch 4/30
152/152 [==============================] - 12s 76ms/step - loss: 0.5096 -
accuracy: 0.8533 - val_loss: 7.5732 - val_accuracy: 0.3481
Epoch 5/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4933 -
accuracy: 0.8568 - val_loss: 7.9239 - val_accuracy: 0.3130
Epoch 6/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4542 -
accuracy: 0.8673 - val_loss: 8.4279 - val_accuracy: 0.3030
Epoch 7/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4386 -
accuracy: 0.8733 - val_loss: 7.2483 - val_accuracy: 0.3609
Epoch 8/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4430 -
accuracy: 0.8662 - val_loss: 8.1701 - val_accuracy: 0.3176
Epoch 9/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4681 -
accuracy: 0.8649 - val_loss: 6.6844 - val_accuracy: 0.4014
Epoch 10/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4546 -
accuracy: 0.8689 - val_loss: 8.5690 - val_accuracy: 0.2991
Epoch 11/30
```

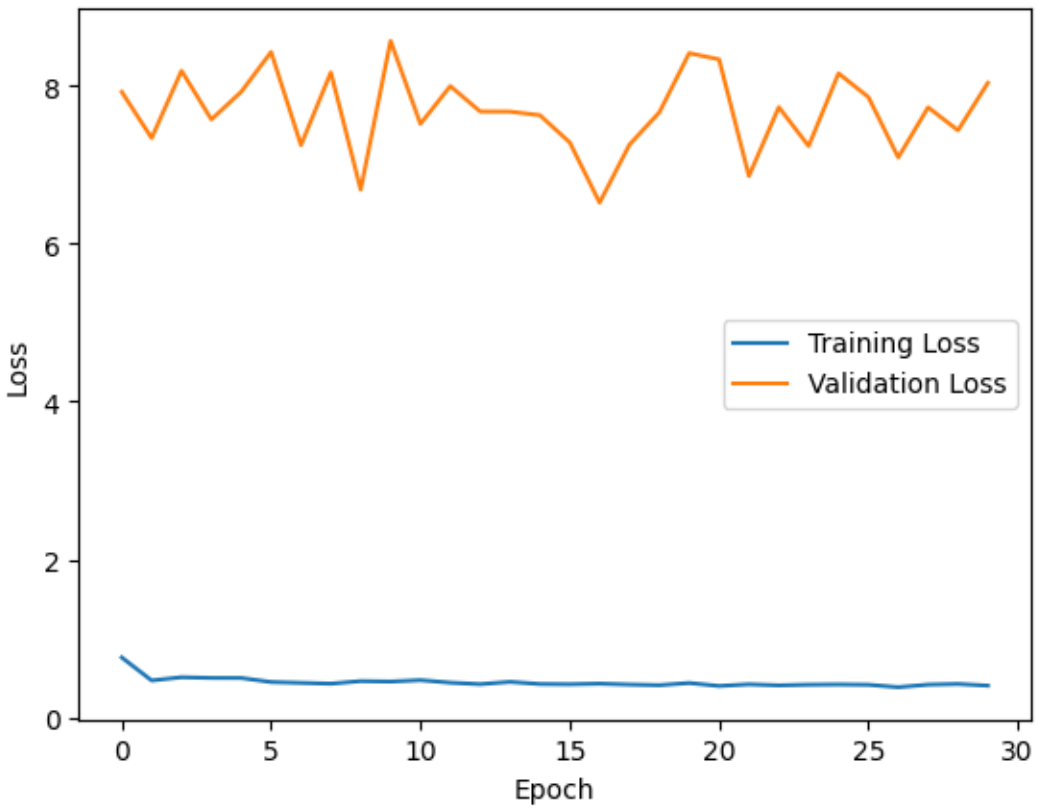
```
152/152 [==============================] - 12s 76ms/step - loss: 0.4617 -
accuracy: 0.8636 - val_loss: 7.5174 - val_accuracy: 0.3407
Epoch 12/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4419 -
accuracy: 0.8721 - val_loss: 7.9964 - val_accuracy: 0.3346
Epoch 13/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4089 -
accuracy: 0.8785 - val_loss: 7.6746 - val_accuracy: 0.3421
Epoch 14/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4447 -
accuracy: 0.8693 - val_loss: 7.6728 - val_accuracy: 0.3414
Epoch 15/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4222 -
accuracy: 0.8768 - val_loss: 7.6273 - val_accuracy: 0.3513
Epoch 16/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4221 -
accuracy: 0.8744 - val_loss: 7.2830 - val_accuracy: 0.3609
Epoch 17/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4296 -
accuracy: 0.8745 - val_loss: 6.5193 - val_accuracy: 0.4128
Epoch 18/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4119 -
accuracy: 0.8779 - val_loss: 7.2549 - val_accuracy: 0.3606
Epoch 19/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4086 -
accuracy: 0.8810 - val_loss: 7.6632 - val_accuracy: 0.3467
Epoch 20/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4386 -
accuracy: 0.8752 - val_loss: 8.4134 - val_accuracy: 0.3147
Epoch 21/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4011 -
accuracy: 0.8869 - val_loss: 8.3344 - val_accuracy: 0.3137
Epoch 22/30
152/152 [==============================] - 12s 76ms/step - loss: 0.3964 -
accuracy: 0.8814 - val_loss: 6.8566 - val_accuracy: 0.3865
Epoch 23/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4094 -
accuracy: 0.8805 - val_loss: 7.7294 - val_accuracy: 0.3300
Epoch 24/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4155 -
accuracy: 0.8754 - val_loss: 7.2366 - val_accuracy: 0.3648
Epoch 25/30
152/152 [==============================] - 12s 76ms/step - loss: 0.4357 -
accuracy: 0.8764 - val_loss: 8.1578 - val_accuracy: 0.3197
Epoch 26/30
152/152 [==============================] - 12s 77ms/step - loss: 0.4122 -
accuracy: 0.8816 - val_loss: 7.8564 - val_accuracy: 0.3275
Epoch 27/30
```

```
152/152 [==============================] - 12s 76ms/step - loss: 0.3856 -
accuracy: 0.8838 - val_loss: 7.0925 - val_accuracy: 0.3787
Epoch 28/30
152/152 [==============================] - 12s 76ms/step - loss: 0.3992 -
accuracy: 0.8822 - val_loss: 7.7271 - val_accuracy: 0.3488
Epoch 29/30
152/152 [==============================] - 12s 79ms/step - loss: 0.4184 -
accuracy: 0.8783 - val_loss: 7.4352 - val_accuracy: 0.3666
Epoch 30/30
152/152 [==============================] - 12s 79ms/step - loss: 0.4148 -
accuracy: 0.8749 - val_loss: 8.0372 - val_accuracy: 0.3346
```

[37]:
```
test_loss, test_acc = model_CNN.evaluate(test_dataset)
print(f"Test Accuracy: {test_acc:.4f}")
```

```
135/135 [==============================] - 2s 17ms/step - loss: 8.3790 -
accuracy: 0.3245
Test Accuracy: 0.3245
```

[38]:
```
plt.plot(history_CNN4.history['loss'], label='Training Loss')
plt.plot(history_CNN4.history['val_loss'], label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```


Training Loss
Validation Loss
Loss
Epoch
0
2
4
6
8
0
5
10
15
20
25
30

[ ]:

[ ]: